ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Marine Technician (MAR TECH)

USQ – Engineering Watchkeeper HFX
(NQual ALPI)

MITE Course IDs:

Residential: 121632
OJPR: 121633

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RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commandant of the Curriculum Control Authority CCA (NFS(P)). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J "Course History".

Revision	Approval Date	Authority
1.0		
3.5	Endorsed - June 2018	CNPTG

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FOREWORD

1. This training document is developed under the authority of the Commander Naval Training and Personnel Group (CNPTG) and in accordance with the following references:

   a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. Annex C – 5555-00379-MAR-TECH dated 15 Dec 16;

   c. NAVORD 4500-0 - Structured Delegation; and

   d. NPTORD 4500-22 – Qualification Standard and Plan Production and Management.

2. This QSP consists of specific Performance Objectives (POs) in Chapter 2 and Lesson Specifications and Enabling Objectives (EOs) in Chapter 4. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA, who will forward to NPTG for approval, as appropriate.

5. The designated TE(s) for this training is indicated below:

TE	Phase	MITE Course ID		NQual
		Residential	OJPR	
NFS(P)	All	121632	121633	APLI
NFS(A)	All	121632	121633	APLI

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Halifax Class Engineering Watchkeeper. This QSP is designed to assist training establishments in developing, conducting, evaluating and validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with NPTORD 4500-22- Qualification Standard and Plan, para 4.5. For this course, a training day consists of 400 minutes.

	Minutes
Actual Training Time (ATT) (Sum of all EOs/POs excluding PC&ECs)	8485
Enabling Checks (ECs)	4800
Performance Checks (PCs)	300
Sub-total	13585 min **33.96 trg days**
Allocation of BAT, TAT, TTT, PT IAW NPTORD 4500-22, Table 5.5	
Base Administrative Time (BAT)	527.55
Training Administrative Time (TAT)	527.55
Training Travel Time (TTT)**	0
Physical Fitness (PT)	1222.65
Sub-total	2277.75 min **5.69 trg days**
Course Duration Calculation (ATT+PCs&ECs+BAT+TTT+TAT+PT) * **Partial days must be rounded up** ** **TTT is intended for travel to and from PT, 50 minutes per PT period**	15862.75min **39.66 trg days**

RESOURCES

3. Identify key personnel and facilities required to conduct the training. Additional training establishment resources can be listed in Annex H if required.

Required Course Staffing					
EO	EO Name	Min Qualification	Min Rank	Qty for Instruction	Qty for Assessment
001.01	Explain the IPMS	Mar Tech – MS IPMS Tech or MAR TECH MS Elec qualified (Ideally IPMS OBTS & BDCS Scenario Building 119167-0006) OR MAR TECH Mech Sr Wkpr/Cert 3E qualified.	MS OR MS/ PO2	1	2
001.02	Explain the Ancillary and Auxiliary Systems				
001.03	Explain the Power Generation and Distribution System				
001.04	Explain the Shafting Arrangements System				
001.05	Explain Main Gearing				
001.06	Explain Propulsion Diesel Engine				
001.07	Explain the Gas Turbine Main Propulsion Engines				
002.01	Operate Machinery Console				
002.02	Operate Auxiliary Systems				
002.03	Operate Ancillary Systems				
002.04	Operate Power Generation and Distribution (PG&D)				
002.05	Operate Gearing & Shafting				
002.06	Operate Main Propulsion				
003.01	Investigate Auxiliary Systems				
003.02	Investigate Ancillary Systems				
003.03	Investigate PG&D				
003.04	Investigate Gearing & Shafting				
003.05	Investigate Main Propulsion				
004.01	Respond to Auxiliary and Ancillary Alarms				
004.02	Respond to PG&D Alarms				
004.03	Respond to Main Propulsion and Gearing and Shafting Alarms				
004.04	Respond to Ship’s Emergencies				
OJPR	On-Board Phase	Chief Eng	CPO2	Nil	1

Critical Training Resources	No of Days
Propulsion Control Operator Trainer (PCOT)	12.34
Machinery Control Room (MCR) Trainer	9

It is possible a Fleet School may not have the necessary equipment or other resources to conduct training as laid out in this QSP. In such cases, alternate methods as documented by Training Deviation Requests (TDRs) may be considered and/or changes in scheduled timings for the course shall be considered.

COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40 – Quantity Control Process for the NTS. Resource implications need to be considered if a course size is above the maximum load.

    a. Maximum: 12;

    b. Minimum: 1; and

    c. Optimum: 12.

MITE

5. Loading. The authority to load members onto this course lies with the Training Establishment (TE). NPTG is to be included as an INFO addressee on all loading messages.

6. Scheduling. Each session of this course shall be scheduled IAW NPTORD 4500-1 – Individual Training and Education Policy and NPTORD 4500-40 Quantity Control Management for the Naval Training System; in MITE by the Training Establishment conducting the training, and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends, and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays.

7. Prerequisites. To be selected for this course the loading authority must ensure that trainees have the following prerequisites:

    a. Interim HFX Class Roundsman (MITE 121028);

    b. USQ Roundsperson HFX (MITE 121178/121179) NQual ALMJ; and

    c. Damage Control System Operator (MITE 120071).

    d. USQS1 Mech (MITE 122019/122969) NQual ALUS or USQS1 Elec (MITE 122020) NQual ALUT qualification.

8. Qualification. Successful completion of this course constitutes the primary means of attaining the Engineering Watchkeeper Halifax Class (NQual: ALPI).

COURSE REPORTING

9. Upon passing this course, trainees shall receive a Course Report (CF 377) and Certificate of Military Achievement (CF 289) completed IAW CMP Instruction 01/06 - Course Reporting and Certificates and NPTORD 4500-51 - Course Reports and Certification.

EVALUATION

10. Program evaluation shall be conducted IAW the guidance provided in A-P9-050-000 CFITES Vol. 11 - Evaluation and Validation Techniques.

VALIDATION

11. This training will be validated every five years CFITES Vol. 8 - Validation of Instructional Programs and based on the task list in Annex B.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

12. PLARs shall be conducted in accordance with NPTORD 4500-27 – PLAR, DAOD 5031-1 - Canadian Forces Military Equivalencies Program and CFITES Vol 12 - Canadian Forces Military Equivalencies Program: Prior Learning Assessment.

SPECIAL INSTRUCTIONS

13. The following are pre-requisites for a legacy trade member to be loaded on the USQ – Engineering Watchkeeper HFX course:

    a. Interim HFX Class Roundsman (MITE 121028);

    b. USQ Roundsperson HFX (MITE 121178/121179) NQual ALMJ;

    c. Damage Control System Operator (MITE 120071); and

    d. One of:

        i. QL5 MAR ENG (MITE 109363/109364, NQual AKTO);

        ii. QL5 HTECH (MITE 108493, NQual AIFF); and

iii. QL5 ETECH (MITE 109328/109329, NQual AJTL)

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

PO 001 EXPLAIN THE OPERATION OF ENGINEERING SYSTEMS

1. Performance. Explain the operation of engineering systems.

2. Conditions.

   a. Given:

      (1) Operational IPMS/ OBTS;

      (2) References; and

      (3) Supervision.

   b. Denied: Assistance.

   c. Environmental: At sea or alongside in any weather condition.

3. Standard. The Engineering Watchkeeper shall, IAW references, explain the operation of engineering systems including:

   a. IPMS;

   b. Ancillary and auxiliary systems;

   c. Power Generation & Distribution (PG&D) system;

   d. Shafting arrangements;

   e. Main Gearing;

   f. Propulsion Diesel Engine (PDE); and

   g. Gas Turbines (GTs) main propulsion engines.

4. References. A1, A2, A3, A4, A5, A6, A7, A8, A9, A10, A11, A12, A13, A14, A15, A16, A17, A18, A19, A20, A21, A22, A23, A30, A31, A32, A33, A34, A35, A36, A37, A38, A39, A40, A41, A42, A43, A44, C1.

5. Task Numbers. AUX001, AUX002, AUX007, AUX008, AUX014, AUX015, AUX026, AUX029, AUX049, AUX050, AUX053, 060, AUX066, AUX067, AUX074, AUX080, AUX086, AUX141, AUX170, AUX171, PGD002, PGD003, PGD004, PGD021, PGD022, PGD025, PGD027, FDC001, FDC002, FDC003, FDC004, FDC012, FDC013, WKP006, WKP007, WKP008, WKP013, WKP023, WKP024, WKP026, WKP028, WKP029, WKP031, WKP032, WKP033, WKP034, WKP035, WKP036, WKP042, WKP043, WKP044, WKP045, WKP046, WKP047, WKP050, WKP052, WKP061, WKP064,

WKP065, WKP066, WKP067, WKP068, WKP069, WKP070, WKP071, WKP072, WKP073, WKP074, WKP075, WKP076, WKP077, WKP078, WKP079, WKP083, WKP084, WKP088, WKP089, WKP090, WKP091, WKP093, WKP095, WKP099, WKP100, WKP101, WKP102, WKP103, WKP105, WKP106.

6. Training Limitations. Nil.

7. Remarks. Nil.

8. Test Details. PO 001 contains 7 homework written assignments for each EO and 2 written PC (covering EO 001.01 to EO 001.03 and EO 001.04 – EO 001.07).

PO 002 OPERATE INTEGRATED PLATFORM MANAGEMENT SYSTEM (IPMS) CONSOLE

1. Performance. Operate IPMS system.

2. Conditions.

   a. Given:

      (1) Ship with applicable equipment and/or system;

      (2) PPE;

      (3) Operational IPMS system;

      (4) Assistance;

      (5) References; and

      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Engineering Watchkeeper shall operate IPMS system IAW refs by:

   a. Operating machinery console;

   b. Operating auxiliary systems;

   c. Leading Collective Operation of ancillary systems;

   d. Operating PG&D system;

   e. Operating gearing and shafting systems;

   f. Operating main propulsion system; and

4. References. Specified in applicable EOs.

5. Task Numbers. CT0350, WT0010, WT0110, IT0380, ALT1520, HT0060, HT0040, HT0050, HT0070, ET0230, DT0120, ATT0920, XT0090, AIT0870, AIT0890, AIT0620, MT0020, ABT0120, DT0090, QT0400, AVT0770, DT0200, DT0180, DT0070, DT0060, CT0060, CT0280, ABT0050, BT0200, ABT0150, ABT0170, RT0010, ABT0200, QT0410, QT0390, QT0420, ABT0210, AKT1090, GT0010, QT0350, ABT0070, ABT0080, CT0270, WT0040, WT0090, IT0510, AMT0160.

6. Training Limitations. Ship availability during the course.

7. Remarks. Nil.

8. Test Details. PO 002 contains one practical PC covering all EO.

PO 003 INVESTIGATE ENGINEERING SYSTEMS

1. Performance. Investigate engineering systems.

2. Conditions.

   a. Given:

      (1) Ship with applicable equipment and/or system;

      (2) PPE;

      (3) Operational IPMS;

      (4) Assistance;

      (5) References; and

      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Engineering Watchkeeper shall investigate engineering systems IAW refs by:

   a. Investigating collective operation of auxiliary systems;

   b. Investigating collective operation of ancillary systems;

   c. Investigating PG&D system;

   d. Investigating operation of gearing and shafting systems; and

   e. Investigating main propulsion system.

4. References. Specified in applicable EOs.

5. Task Numbers. ALT2200, AHT0860, AHT0890, AHT0840, AHT0830, CT0130, AVT1310, AVT1340, AVT1280, AVT1300, AVT1290, AVT1360, AVT1330, AVT1320, AVT1350, AHT0850, AHT0870, AHT0880, AVT1430, AVT1450, BT0050, BT0060, CT0120, AHT0820, CT0160, CT0150, ALT2170, ALT2180, CT0430, CT0140.

6. Training Limitations. Ship availability during the course.

7. Remarks. Nil.

8. Test Details. PO 003 contains no EC but PO 003 material will be assessed under EC 004.01, EC 004.02 and EC 004.04 (all practical assessments). Material will be also assessed under PC 004.

PO 004 RESPOND TO ALARMS AND EMERGENCIES

1. Performance. Respond to alarms and emergencies.

2. Conditions.

   a. Given:

      (1) Ship with applicable equipment and/or system;

      (2) PPE;

      (3) Operational IPMS;

      (4) Assistance;

      (5) References; and

      (6) Supervision.

   b. Denied: Nil.

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Engineering Watchkeeper shall respond to alarms and emergencies IAW refs by:

   a. Responding to auxiliary and ancillary systems alarms;

   b. Responding to PG&D alarms;

   c. Responding to main propulsion and gearing and shafting systems alarms; and

   d. Responding to ship's engineering emergencies while operating the console.

4. References. Specified in applicable EOs.

5. Task Numbers. JT0020, JT0040, CT0330, CT0050, JT0030, CT0030, PT0010, CT0340, CT0310, CT0320, CT0010, CT0020, CT0040, GT0010, WT0020, IT0470.

6. Training Limitations. Ship availability during the course.

7. Remarks. Nil.

9. Test Details. PO 004 contains 3 practical EC (first EC covers EO 003.01, 003.02 and 004.01, second EC covers EO 003.03 and 004.02, and third EC covers EO 003.04, 003.05, 004.03 and 004.04) and one practical OJPR PC with an oral board as final assessment.

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY NFS(P)

GENERAL

1. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this QSP and the standard listed must be met to achieve this qualification. PCs will be administered in the form of practical examinations and the trainee must obtain a pass in all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative tests that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are designed to give feedback to both the instructor(s) and trainees. Marks for ECs are not calculated in the trainee's final mark.

3. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be done IAW CFITES Vol. 7 – Evaluation of Learners.

5. Senior instructor or delegated staff provides quality control by administering and scoring all Performance Checks. Any PC not administered or invigilated by staff will be administered by a non-involved instructor designated by the senior instructor. Enabling Checks are the responsibility of the Instructional Staff.

PROGRESS MONITORING

6. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW CFITES Vol 6 - Conduct of Instruction Programs and TE SOs or SOPs.

7. Continuous monitoring of candidate progress is required on this course in order to provide the following:

    a. Early warning of trainee difficulties;

    b. Feedback on the effectiveness of training;

    c. Information for the TRB or PRB (if required); and

d. Information to assist in completion of the CF 377 Course Report.

8. Trainees experiencing difficulty in any area of performance are to be counselled WRT the nature of their shortcomings and suggested corrective action IAW Training Establishment SOs or SOPs.

9. PC failures and supplemental PC failures are to be handled IAW Training Establishment SOs or SOPs. Supplemental PCs shall not be given prior to 24 hours having elapsed since trainee was informed of their PC failure.

TRAINING REVIEW BOARD

10. TRBs shall be convened and conducted IAW with NPTORD 4500-54 - Training Review Board Process.

COURSE REPORTS

11. Course reports (CF 377) shall be prepared and distributed IAW CFAO 26-12, NPTORD 4500-51 - Course Reporting and Certificates.

12. Trainees will be graded pass or fail. IAW NPTORD 4500-51 - Course Reporting and Certificates, students will not be ranked.

13. Where necessary, the narrative will indicate trainee's strengths and weaknesses based on PC results. Training limitations and/or incomplete training must also be included in the course report. Record of trainee counselling is to be maintained by the training establishment.

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY NFS(P)

PO 001 Explain the Operation of Engineering Systems.

EO 001.01 Explain the IPMS.

1. Performance. Explain the IPMS.

2. Conditions.

   a. Given:

      (1) Operational IPMS/OBTS;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied: Nil.

   c. Environmental: Classroom/Trainer.

3. Standard. The trainee shall, IAW references, explain the IPMS including:

   a. Architecture

      (1) Network Configuration;

      (2) Human Machine Interfaces (HMIs); and

      (3) Virtual Remote Terminal Units (VRTUs);

   b. Operation

      (1) Station In Control (SICs);

      (2) Overrides;

      (3) Alarms, Warnings and Errors;

   (4) Bypass Options and Tests;

   (5) Propulsion Control Modes; and

   (6) Drive Modes.

4. Time.

   a. 7 x 50 min periods; and

   b. EC review 1 x 50 min periods.

5. Method / Media.

   a. Lecture;

   b. Demonstration;

   c. Ship tour.

6. References. A2, A15, A21, A22, A23, A30, A36.

7. Training Aids.

   a. Power Point presentations;

   b. System drawings;

   c. MCRT/Platform Control Operational Trainer (PCOT); and

   d. Ship.

8. Training Limitations. Ship availability.

9. Test Details. EC 001.01 will be a homework assignment to be corrected during class time. EO 001.01 will be tested in PC 001.03.

10. Remarks. OJT not included in time calculations.

11. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1							
Network Configuration: 1. Location, function, general overview; 2. Distributed Ethernet LAN (DREL); 3. Network Switches; and 4. LAN Drops/Access Points.		50			Lecture/ demo	A15, A21, A22, A23	
TP2							
HMIs: 1. Location, function, general overview; 2. MCC, SC, MEC, BCC, DC1; 3. LOP;		50			Lecture/ demo	A15, A21, A22, A23	

4. LSD; 5. POU; and EHM, OBTS.						
TP3						
VRTUs: 1. Location, function, general overview; 2. Schedules; and ASC, PSC, ECM.	25			Lecture/ demo	A15, A21, A22, A23	
TP4						
SICs: 1. General overview; 2. Hierarchy; and Role attribution.	25			Lecture/ demo	A15, A21, A22, A23	
TP5						
Overrides: 1. General overview; 2. Battle override; and GT overrides.	25			Lecture/ demo	A2, A5, A7, A9, A15, A16, A21, A22, A23	
TP6						
Alarms, Warnings and Errors: 1. General overview; 2. Recognition and indications; and 3. Event Viewer Management.	25			Lecture/ demo	A2, A15, A21, A22, A23	
TP7						
Bypass Options and Tests: 1. General overview; 2. PDE bypasses; 3. GT bypasses; and GT Tests.	50			Lecture/ demo	A2, A5, A7, A9, A15, A16, A21, A22, A23,	

					A30, A36	
TP8						
Propulsion Control Modes: 1. General overview; 2. Automatic; 3. Semi-automatic; and Manual.	50			Lecture/ demo	A2, A9, A15, A21, A22, A23, A30	
TP9						
Drive Modes: 1. General overview; 2. Normal; and Abnormal.	50			Lecture/ demo	A2, A9, A15, A21, A22, A23, A30	

Enabling check - EC 001.01 Explain the IPMS

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 001.01 will be a written homework assessment based on content from EO 001.01 and reviewed during class time. EO 001.01 will be tested as part of PC001.03.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.01 will be a written homework assessment based on content from EO 001.01.

2. Read each question carefully, and all responses fully, before answering the question.

3. Written homework will be reviewed and verified in a group discussion setting.

4. Do you have any questions?

PART III – ENABLING CHECK NOTES

1. Instructor to verify homework completed prior to discussion group.

2. Trainees to ensure accuracy of answers based on group discussion led by instructor.

3. Written homework to become self-study aid for PC 001.03 and PC 004.

PART IV – ENABLING CHECK LIST

N/A.

EO 001.02 Explain Ancillary and Auxiliary Systems.

1. Performance. Explain ancillary and auxiliary systems.

2. Conditions.

   a. Given:

      (1) Operational IPMS/OBTS;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied: Nil.

   c. Environmental: Classroom/Trainer.

3. Standard. The trainee shall, IAW references, explain ancillary and auxiliary systems(HFX Class) including:

   a. Controllable Reversible Pitch Propeller (CRPP);

   b. Fuel Oil Systems (FOS);

   c. Main Lube Oil (MLO);

   d. Steering Gear (STG);

   e. Sea Water Cooling Systems (SWC); and

   f. High Pressure and Low Pressure Air systems (HP and LP).

4. Time.

   a. 20 x 50 min periods; and

   b. EC review – 1 x 50 min period.

5. Method / Media.

   a. Lecture;

   b. Demonstration – time captured during lecture periods; and

   c. Ship tour.

6. References. A1, A2, A3, A4, A7, A8, A9, A10, A13, A14, A21, A22, A23, A30, A32, A34, A35, A37, A38, A39, A42, C1.

7. Training Aids.

   a. Power Point presentations;

   b. System drawings;

   c. MCRT/PCOT; and

   d. Ship.

8. Training Limitations. Ship availability.

9. Test Details. EC 001.02 will be a homework assignment to be corrected during class time. EO 001.02 will be tested in PC 001.3

10. Remarks. OJT not included in time calculations.

11. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1							
CRPP: 1. Location, function, general overview. 2. Pumps: a. Main pump; b. Standby pumps; c. Header pump; and d. Hand pump. 3. Tanks: a. Storage tank; and b. Header tank. 4. Cooler; 5. Valves: a. Balanced piston; b. Relief; and c. Cross-connects. 6. Actuating Unit; 7. Hub assembly;		200			Lecture/ demo	A2, A9, A21, A22, A23, A30, A35, A37	

8. Filters; 9. Oil tubes: a. Ahead/Astern; and b. Sealing. 10. Modes of Operation: a. Normal Operation; and b. Emergency Operation. 11. Parameters: a. Trips; b. Alarms; and 12. Warnings.						
TP2						
Fuel Oil Systems – Fuel Oil Services (FOS): 1. Location, function, general overview; 2. Tanks: a. Service; b. Settling; c. Day; d. Fuel recovery; and e. Emergency fuel. 3. Pumps: a. Boost; and b. Hand. 4. Filters and Strainers: a. Fuel Water Separator; and b. Suction Strainers. 5. End Users: a. GTs; b. PDE; c. Diesel generators; d. Steam generator; e. Diesel Driven Fire Pump (DDFP); and f. Rigid Inflatable Boat (RIB).	100			Lecture/ demo	A2, A8, A9, A13, A14, A21, A22, A23, A30	

6. Modes of operation: a. Normal; and b. Emergency. 7. Parameters: a. Alarms; b. Trips; and Warnings.						
TP3						
Fuel Oil Systems – Fuel Oil Filling and Transfer (FOT): 1. Location, function, general overview; 2. Tanks: a. Storage; and b. Sea water ballast. 3. Pumps; 4. Fuel centrifuge; 5. Fuel management; 6. Modes of operation: a. Normal; and b. Emergency; 7. Parameters; a. Alarms; b. Trips; and c. Warnings. Liquid Level Management (LLM).	50			Lecture/ demo	A2, A8, A9, A13, A14, A21, A22, A23, A30	
TP4						
MLO: 1. Location, function, general overview; 2. Tanks; 3. Pumps; a. Electrical; and b. Air driven. 4. Filters; 5. Cooler;	200			Lecture/ demo	A2, A7, A9, A21, A22, A23, A30, A32, A34, C1	

6. Valves; a. Relief; b. Regulating; and c. Thermostatic control. 7. Lube Oil Storage and Conditioning Assembly (LOSCA); 8. Lube oil filling and transfer: a. Tanks; b. Heaters; and c. Separators. 9. Modes of operation: a. Normal; and b. Emergency. 10. Parameters: a. Alarms; b. Trips; and c. Warnings.						
TP5						
Steering: 1. Location, function, general overview; 2. Pumps: a. Main; b. Centering; and c. Hand. 3. Tanks; 4. Filters; 5. Valves: a. Shock; and b. Directional Control (DCVs). 6. Accumulators; 7. Modes of operation: a. Normal: (1) Automatic Follow Up (AFU); (2) Follow Up (FU); and	200			Lecture/ demo	A2, A9, A21, A22, A23, A30, A38, A39	

(3) Non Follow Up (NFU). b. Emergency: (1) Centering; and (2) Locking. 8. Parameters: a. Alarms; b. Trips; and Warnings.						
TP6						
Sea Water Cooling Systems – Sea Water Services (SWS): 1. Location, function, general overview; 2. End users: a. Emergency ASC supply; b. Fitted firefighting systems; and c. Emergency and supplementary cooling. 3. Modes of operation: a. Normal; and b. Emergency. 4. Parameters: a. Alarms; b. Trips; and c. Warnings. 5. Firemain subsystem.	50			Lecture/ demo	A1, A2, A3, A9, A10, A21, A22, A23, A28, A42	
TP7						
Sea Water Cooling Systems – Main and Auxiliary Sea Water Circulation (MSC and ASC): 1. Location, function, general overview; 2. Pumps: a. MSC; and b. ASC. 3. Valves: a. 32 mm;	100			Lecture/ demo	A1, A2, A3, A7, A9, A10, A21, A22, A23, A28, A42	

b. 150 mm; and c. Overboard regulating. 4. Modes of operation: a. Normal; and b. Emergency. 5. Parameters: a. Alarms; b. Trips; and c. Warnings.						
TP8						
Air systems – High Pressure (HP): 1. Location, function, general overview; 2. End users: a. Clutches; b. Power turbine brake; and c. Clutch control cabinet; d. Reducing stations. 3. Modes of operation: a. Normal; and b. Emergency. 4. Parameters: a. Alarms; b. Trips; and c. Warnings.	50			Lecture/ demo	A2, A3, A4, A7, A9, A21, A22, A23, A30, A42	
TP9						
Air systems – Low Pressure (LP): 1. Location, function, general overview; 2. Control Air Stations (CAS); 3. Modes of operation: a. Normal; and b. Emergency. 4. Parameters: a. Alarms; b. Trips; and	50			Lecture/ demo	A2, A3, A4, A7, A9, A21, A22, A23, A30, A42	

c. Warnings.						

Enabling check - EC 001.02 Explain Ancillary and Auxiliary Systems.

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 001.002 will be a homework assessment to be reviewed during class time. EO 001.02 will be tested under PC 001.03.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.02 will be a written homework assessment based on content from EO 001.02.

2. Read each question carefully, and all responses fully, before answering the question.

3. Written homework will be reviewed and verified in a group discussion setting.

4. Do you have any questions?

PART III – ENABLING CHECK NOTES

1. Instructor to verify homework completed prior to discussion group.

2. Trainees to ensure accuracy of answers based on group discussion led by instructor.

3. Written homework to become self-study aid for PC 001.03 and PC 004.

PART IV – ENABLING CHECK LIST

N/A.

EO 001.03 Explain the Power Generation and Distribution System (PG&D).

1. Performance. Explain the PG&D system.

2. Conditions.

   a. Given:

      (1) Operational IPMS/OBTS;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied: Nil.

   c. Environmental: Classroom/Trainer.

3. Standard. The trainee shall, IAW references, explain the PG&D system including:

   a. Motoren Werke Mannheim (MWM) diesel generator set;

   b. Caterpillar diesel generator set (DG830); and

   c. Electrical distribution system.

4. Time.

   a. 7 x 50 min periods; and

   b. EC Review – 1 x 50 min periods.

5. Method / Media.

   a. Lecture;

   b. Demonstration; and

   c. Ship tour.

6. References. A2, A4, A15, A21, A22, A23, A30, A40, A41, A44.

7. Training Aids.

   a. Power Point presentations;

   b. System drawings;

   c. MCRT/PCOT; and

d. Ship.

8. Training Limitations. Ship availability.

9. Test Details. EC 001.03 will be a homework assignment to be corrected during class time. EO 001.03 will be tested in PC 001.03

10. Remarks. OJT not included in time calculations.

11. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1							
MWM diesel generator set: 1. Location, function, general overview; 2. Modes of operation: a. Locations for start and stop; b. Emergency starts and stops; and c. Fail starts and stops. 3. Parameters: a. Alarms; b. Trips; and c. Warnings.		50			Lecture/ demo	A4, A13, A15, A21, A22, A23, A40, A41, A42, A44	
TP2							
CAT diesel generator set: 1. Location, function, general overview; 2. Modes of operation: a. Locations for start and stop; b. Emergency starts and stops; and c. Fail starts and stops. 3. Parameters: a. Alarms; b. Trips; and c. Warnings.		50			Lecture/ demo	A4, A13, A15, A21, A22, A23, A40, A41, A42, A44	

TP3						
Electrical distribution system: 1. Location, function, general overview; 2. Switchboards: a. Electrical Power Control System (EPCS); and b. Configuration. 3. Power supplies and loads: a. Vital loads; b. Semi-vital loads; c. Non-vital loads; d. Casualty power; e. Automatic Change Over Switch (ACOS); f. Manual Change Over Switch (MCOS); g. Static Frequency Converter (SFC); and h. Uninterruptable Power Supplies (UPS). 4. Modes of operation: a. Automatic; b. Manual; c. Manual permissive; and d. Emergency operation. 5. Parameters: a. Alarms; b. Trips; and c. Warnings.	200			Lecture/ demo/ OJT	A15, A21, A22, A23	

Enabling check - EC 001.03 Explain the Power Generation and Distribution System (PG&D).

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 001.03 will be a written homework assessment based on content from EO 001.03 and reviewed during class time. EO 001.03 will be tested under PC 001.03.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.03 will be a written homework assessment based on content from EO 001.03.

2. Read each question, and all responses carefully, before answering the question.

3. Written homework will be reviewed and verified in a group discussion setting.

4. Do you have any questions?

PART III – ENABLING CHECK NOTES

1. Instructor to verify homework completed prior to discussion group.

2. Trainees to ensure accuracy of answers based on group discussion led by instructor.

3. Written homework to become self-study aid for PC 001.03 and PC 004.

PART IV – ENABLING CHECK LIST

N/A.

PERFORMANCE CHECK - PC 001.03

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. PC 001.03 will be a written test based on content from EO 001.01 through EO 001.03.

3. Pass mark will be 60%.

4. Time required: 2 x 50 min periods.

5. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 001.03 will be a written test based on content from EO 001.01 through EO 001.03.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will not be provided.

9. Pass mark will be 60%.

10. You will have 2 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Examination provided by NFS(P) MSE Division

PART IV – EDUCATIONAL CHECK LIST

N/A.

EO 001.04 Explain Shafting Arrangements.

1. Performance. Explain shafting arrangements.

2. Conditions.

    a. Given:

        (1) Operational IPMS/OBTS;

        (2) Schematics and illustrations from naval and manufacturers' manuals; and

        (3) References.

    b. Denied: Nil.

    c. Environmental: Classroom/Trainer.

3. Standard. The trainee shall, IAW references, explain shafting arrangements including:

    a. Shafting system components; and

    b. Modes of operation.

4. Time.

    a. 5 x 50 min periods; and

    b. EC – homework assignment to be corrected in class.

5. Method / Media.

    a. Lecture;

    b. Demonstration; and

    c. Ship tour.

6. References. A2, A9, A17, A18, A19, A20, A21, A22, A23, A30, A35.

7. Training Aids.

    a. Power Point presentations;

    b. System drawings;

    c. MCRT/PCOT; and

    d. Ship.

8. Training Limitations. Ship availability.

9. Test Details. EC 001.04 will be a homework assignment to be corrected during class time. EO 001.04 will be tested in PC 001.07.

10. Remarks. Nil.

11. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1							
Shafting system components – shaft segments: 1. Location, function, and general overview: a. Thrust shaft; b. Intermediate 2 shaft; c. Actuating shaft; d. Intermediate 1 shaft; e. Stern tube shaft; and f. Propeller shaft.		50			Lecture/ demo	A2, A7, A9, A17, A18, A19, A20, A21, A22, A23, A30, A35	
TP2							
Shaft system components - couplings: 1. Flexible; 2. Loose; and 3. Flange.		50			Lecture/ demo	A2, A7, A9, A17, A18, A19, A20, A21, A22, A23, A30, A35	
TP3							
Shaft system components – supporting features: 1. Combined thrust and plummer block; 2. Plummer bearings: a. Baseline; and b. Centerline.		100			Lecture/ demo	A2, A7, A9, A17, A18, A19,	

3. Bulkhead seals; 4. Torsion meter; 5. CRPP actuating unit; 6. Shaft grounding: a. Active; and b. Passive. 7. Stern seal; 8. 'A' brackets and bearings: a. Intermediate; and b. Main. 9. Eddy plate; 10. Rope guards; 11. Propeller hub; and 12. Exterior shaft cladding.					A20, A21, A22, A23, A30, A35	
TP4						
Modes of operation: 1. Normal; 2. Emergency; and 3. Parameters: a. Alarms; b. Trips; and c. Warnings.	50			Lecture/ demo	A2, A7, A9, A17, A18, A19, A20, A21, A22, A23, A30, A35	

Enabling check - EC 001.04 - Explain Shafting Arrangements.

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 001.04 will be a written homework assessment based on content from EO 001.04 and reviewed during class time. EO001.04 will be tested under PC001.07.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.04 will be a written homework assessment based on content from EO 001.04.

2. Read each question carefully, and all responses fully, before answering the question.

3. Written homework will be reviewed and verified in a group discussion setting.

4. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Instructor to verify homework completed prior to discussion group.

2. Trainees to ensure accuracy of answers based on group discussion led by instructor.

3. Written homework to become self-study aid for PC.

PART III – ENABLING CHECK NOTES

Give guidance for the development of the assessment.

PART IV – ENABLING CHECK LIST

N/A.

EO 001.05 Explain Main Gearing.

1. Performance. Explain main gearing.

2. Conditions.

   a. Given:

      (1) Operational IPMS/OBTS;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied: Nil.

   c. Environmental: Classroom/Trainer.

3. Standard. The trainee shall, IAW references, explain main gearing including:

   a. Physical description and components; and

   b. Modes of operation.

4. Time.

   a. 6 x 50 min periods; and

   b. EC – homework assignment to be corrected in class.

5. Method / Media.

   a. Lecture;

   b. Demonstration; and

   c. Ship tour.

6. References. A2, A4, A7, A9, A21, A22, A23, A30, A32, C1.

7. Training Aids.

   a. Power Point presentations;

   b. System drawings;

   c. MCRT/PCOT; and

   d. Ship.

8. Training Limitations. Ship availability.

9. Test Details. EC 001.05 will be a homework assignment to be corrected during class time. EO 001.05 will be tested in PC 001.07

10. Remarks. Nil.

11. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1							
Physical description and components – location, function and general overview: 1. Main rafting layout; 2. Gearing arrangement: a. Main gearing; b. Cross connect gearing; c. Construction and design; d. Reduction ratios; e. Pinions; f. Idlers; g. Input shafts; h. Quill shafts; i. Primary gear wheel; and j. Main gear wheel. 3. Sealing components: a. Labyrinth; b. Stuffing box; c. Oil muff; and d. Inflatable. 4. Main sea water circulation pump drive; 5. Turning gear; 6. Shaft lock; and 7. Flexible diaphragm coupling.		100			Lecture/ demo	A2, A4, A7, A9, A21, A22, A23, A30, A32	
TP2							
Physical description and components - clutches: 1. Synchronous Self Shifting (SSS) clutch; 2. Friction clutches: a. Cross connect; and b. PDE. 3. Clutch controls.		100			Lecture/ demo	A2, A4, A7, A9, A21, A22, A23,	

					A30, A32	
TP3						
Physical description and components – bearings: 1. Axial locating; 2. Journal; and 3. Lubrication.	50			Lecture/ demo	A2, A4, A7, A9, A21, A22, A23, A30, A32	
TP4						
Modes of operation: 1. Normal; 2. Emergency; 3. Monitoring: a. Resistive Temperature Device (RTD); b. Signal Conditioning Unit (SCU); and c. Speed sensing element. 4. Parameters: a. Alarms; b. Trips; and c. Warnings. 5. Bearing Flow Chart	50			Lecture/ demo	A2, A4, A7, A9, A21, A22, A23, A30, A32	

Enabling check - EC 001.05 - Explain Main Gearing.

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 001.05 will be a written homework assessment based on content from EO 001.05 and reviewed during class time. EO 001.05 will be tested under PC 001.07.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.05 will be a written homework assessment based on content from EO 001.05.

2. Read each question carefully, and all responses fully, before answering the question.

3. Written homework will be reviewed and verified in a group discussion setting.

4. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Instructor to verify homework completed prior to discussion group.

2. Trainees to ensure accuracy of answers based on group discussion led by instructor.

3. Written homework to become self-study aid for PC.

PART III – ENABLING CHECK NOTES

Give guidance for the development of the assessment.

PART IV – ENABLING CHECK LIST

N/A.

EO 001.06 Explain the Propulsion Diesel Engine (PDE).

1. Performance. Explain the PDE.

2. Conditions.

   a. Given:

      (1) Operational IPMS/OBTS;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied: Nil.

   c. Environmental: Classroom/Trainer.

3. Standard. The trainee shall, IAW references, explain the PDE including:

   a. Physical description and components; and

   b. Modes of operation.

4. Time.

   a. 9 x 50 min periods; and

   b. EC – homework assignment to be corrected in class.

5. Method / Media.

   a. Lecture;

   b. Demonstration; and

   c. Ship tour.

6. References. A2, A3, A4, A7, A9, A13, A21, A22, A23, A30, A31, A36, A42.

7. Training Aids.

   a. Power Point presentations;

   b. System drawings;

   c. MCRT/PCOT; and

d. Ship.

8. Training Limitations. Ship availability.

9. Test Details. EC 001.06 will be a homework assignment to be corrected during class time. EO 001.06 will be tested in PC 001.07.

10. Remarks. Nil.

11. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Physical description and components to include:							
1. Location, function and general overview; 2. Local Control Cubicle (LCC); 3. Fuel system; 4. Lube oil system; 5. Turbo water wash system; 6. Air systems: a. Air start; b. Turbo sealing; c. Control air; and d. Safety slide valve. 7. Intake and exhaust: a. Bypasses; and b. Demisters. 8. Water cooling systems: a. Seawater; and b. Freshwater. 9. Barring gear; 10. Governors: a. Electronic; and Pneumatic.		450 Min			A2, A3, A4, A7, A9, A13, A21, A22, A23, A30, A36	A2, A3, A4, A7, A9, A13, A21, A22, A23, A30, A36	
TP2 Modes of operation to include:							
1. Normal; 2. Emergency; and 3. Parameters: a. Alarms; b. Trips; and c. Warnings.		100			IL DP	A2, A3, A4, A7, A9, A13, A21, A22, A23, A30, A36	

Enabling check - EC 001.06 Explain the Propulsion Diesel Engine (PDE).

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 001.06 will be a written homework assessment based on content from EO 001.06 and reviewed during class time. EO 001.06 will be tested under PC 001.07.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.06 will be a written homework assessment based on content from EO 001.06.

2. Read each question carefully, and all responses fully, before answering the question.

3. Written homework will be reviewed and verified in a group discussion setting.

4. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Instructor to verify homework completed prior to discussion group.

2. Trainees to ensure accuracy of answers based on group discussion led by instructor.

3. Written homework to become self-study aid for PC.

PART III – ENABLING CHECK NOTES

Give guidance for the development of the assessment.

PART IV – ENABLING CHECK LIST

N/A

EO 001.07 Explain the General Electric LM 2500 gas turbine propulsion system.

1. Performance. Explain the General Electric LM 2500 gas turbine propulsion system.

2. Conditions.

   a. Given:

      (1) Operational IPMS/OBTS;

      (2) Schematics and illustrations from naval and manufacturers' manuals; and

      (3) References.

   b. Denied: Nil.

   c. Environmental: Classroom/Trainer.

3. Standard. The trainee shall, IAW references, explain the General Electric LM 2500 gas turbine propulsion system including:

   a. Physical description and components; and

   b. Modes of operation.

4. Time.

   a. 13 x 50 min periods;

   b. EC – 1 x 50 min;

   c. Ship tour – 4 x 50 min periods; and

   d. PC Review – 3 x 50 min periods (2 before, 1 after).

5. Method / Media.

   a. Lecture;

   b. Demonstration; and

   c. Ship tour.

6. References. A2, A4, A5, A6, A7, A9, A11, A12, A16, A21, A22, A23, A30, A33, A43.

7. Training Aids.

   a. Power Point presentations;

b. System drawings;

c. MCRT/PCOT; and

d. Ship.

8. Training Limitations. Ship availability.

9. Test Details. EC 001.07 will be a written homework assessment to be reviewed during class. PC 001.07 will include information from EO 001.04 through EO 001.07.

10. Remarks. The ship tour can be scheduled during EO 003.04. Instructors should take trainees through AMRs, Engine Rooms, and MCR to discuss what the responsibilities of the operator are.

11. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Physical description and components to include:							
1. 1 .Location, function and general overview: a. Compressor; b. Combustion section; c. HP turbine; d. Power turbine; and e. Accessory gearbox. 2. Hydraulic Start System (HSS); 3. Intakes/Uptakes; 4. Fuel system; 5. Water wash; 6. Bleed air; 7. Enclosure and mounting arrangement; 8. Anti-icing; 9. Synthetic lube oil; and 10. Ignition.		550 Min			IL/ D	A2, A4, A5, A6, A7, A9, A11, A12, A13, A16, A21, A22, A23, A30, A33, A43	
TP2 Modes of operation to include:							

4. Normal; 5. Emergency; and 6. Parameters: a. Alarms; b. Trips; and c. Warnings.	100		IL DP	A2, A4, A5, A6, A7, A9, A11, A12, A13, A16, A21, A22, A23, A30, A33, A43	

Enabling check - EC 001.07 Explain the General Electric LM 2500 gas turbine propulsion system.

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

1. EC 001.07 will be a written homework assessment based on content from EO 001.07 and reviewed during class time.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.07 will be a written homework assessment based on content from EO 001.07 and reviewed during class time. The practical component of EO 001.07 is contained in PC 001.07.

2. Read each question carefully before answering the question.

3. Written homework will be reviewed and verified in a group discussion setting.

4. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Instructor to verify homework completed prior to discussion group.

2. Trainees to ensure accuracy of answers based on group discussion led by instructor.

3. Written homework to become self-study aid for PC.

PART III – ENABLING CHECK NOTES

Give guidance for the development of the assessment.

PART IV – ENABLING CHECK LIST

N/A.

PERFORMANCE CHECK – PC 001.07

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions
1. Classroom.

Arrangements and Procedures
1. PC 001.07 will be a written test based on content from EO 001.04 through EO 001.07.
2. Pass mark will be 60%.
3. Time required: 2 x 50 min periods.
4. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 001.07 will be a written test based on content from EO 001.04 through EO 001.07
2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.
3. Do not open the booklet until you are told to do so.
4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.
5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).
6. Read each question carefully, and all responses fully, before answering the question.
7. If you have any questions during the exam, raise your hand.
8. References will not be provided.
9. Pass mark will be 60%.
10. You will have 2 x 50 min periods to complete this exam.
11. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Examination provided by NFS(P) MSE Div.

PO 002 OPERATE INTEGRATED PLATFORM MANAGEMENT SYSTEM (IPMS) CONSOLE

EO 002.01 Operate Machinery Control Console

1. Performance. Operate machinery control console.

2. Conditions.

   a. Given:

      (1) Operational IPMS;

      (2) Assistance;

      (3) References; and

      (4) Supervision.

   b. Denied: Nil.

   c. Environmental:

      (1) PCOT; and

      (2) MCT trainer.

3. Standard. The Engineering Watchkeeper shall independently operate the machinery control console, IAW refs by:

   a. Supervising engineering pre-sail checks (CT0350);

   b. Operating IPMS man machinery interface (HT0040) to include:

      (1) Operating IPMS applications (HT0060);

      (2) Operating IPMS EHM (HT0070);

      (3) Operating IPMS software utilities (HT0050);

      (4) Testing components with built-in test equipment (BITE) (ALT1520)

      (5) Investigating system operation logs (AKT1090); and

      (6) Interpreting drawings/schematics (AMT0160).

   c. Managing damage control conditions (IT0380) to include Monitoring of damage control system (IT0510).

4. Time Required.

   a. EO – 9 x 50 minute periods.

5. Instructional Methods.

   a. Interactive lecture; and

   b. Demonstration.

6. Training Aids.

   a. Ship's official "plate" drawings will be required to aid in comparing the drawings/schematics TP; and

   b. Engineering Watchkeeper Handouts presently located on Marine Systems Division Sharepoint site.

7. Training Limitations. Ship availability.

8. Test Details. Trainees will be provided feedback throughout the EO. No formal EC or PC is included in this EO, instead the material will be tested in PC 002.

9. References. A10, A21, A22, A23, A30, A31, A46, A47, B1.

10. Remarks. Nil.

11. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Supervise engineering pre-sail checks to include:							
a. Supervising engineering pre-sail checks by: (1) Explaining Notice for Power; 1. Duties of the flash-up watch; 2. Pre-watch rounds; and 3. Procedures for flashing up equipment.		50			IL	A30 A31	
TP2 Operate IPMS main machinery interface to include:							
a. Operating IPMS applications by: (1) Navigating through pages; and (2) Utilizing maintenance pages		50	50		IL DP	A21 A22 A23	
b. Operating IPMS EHM by:							

(1) Analyze trending.						
c. Operating IPMS software utilities						
d. Testing components with built-in test equipment						
e. Investigate system operation logs by: (1) Reviewing data captured in IPMS logs.						
f. Interpreting drawings/schematics by: (1) Understanding the IPMS interface; and (2) Relating the IPMS schematics to the ships official drawings.	25	25		IL DP		The ship's "plate" drawings will aid in comparing the IPMS pages with the ship's systems.
TP3 Manage damage control conditions to include:						
a. Managing damage control conditions by: (1) Understanding appropriate damage control conditions; (2) Maintaining control over the watertight integrity board; (3) Managing damage control roundspersons during ship evolutions, i.e. Specials; and (4) Informing command of the current status of the ship's watertight integrity.	50	100	100	IL DP	A10 A30 A46 A47 B1	
b. Monitoring of damage control system by: (1) Reviewing the use of the Battle Damage Control System (BDCS); and (2) Understanding how to activate the following damage control systems: i. Firemain; ii. Halon; iii. Aqueous Film Forming Foam (AFFF); iv. Fine Water Spray (FWS); v. Prewet; and						

vi. Automatic water spray (Magazine). (3) Understanding damage control alarms, to include: i. Verbal alarms; and ii. Automatic alarms.						

EO 002.02 Operate Auxiliary Systems

1. Performance. Operate auxiliary systems.

2. Conditions.

   a. Given:

      (1) Operational IPMS;

      (2) Assistance;

      (3) References; and

      (4) Supervision.

   b. Denied: Nil.

   c. Environmental:

      (1) PCOT; and

      (2) MCR Trainer.

3. Standard. The Engineering Watchkeeper shall independently operate auxiliary systems, IAW refs by:

   a. Leading collective operation of auxiliary systems (ET0230) to include:

      (1) Operating steering system remotely (XT0090);

      (2) Operating fuel oil filling/transfer system remotely (DT0120);

      (3) Operating fluid piping system (ATT0920);

      (4) Testing auxiliary seawater circulation automatic sequence function (AIT0870); and

      (5) Testing auxiliary seawater circulation system emergency supply change over function remotely (AIT0890);

   b. Monitoring ship's stability (WT0090) by:

      (1) Assessing ship's stability (WT0040);

      (2) Monitoring liquid level management system (CT0270);

      (3) Adjusting ship's stability (WT0010); and

(4) Recording ship's stability variables (WT0110).

4. Time Required.

a. EO – 9 x 50 minute periods.

5. Instructional Methods.

a. Interactive lecture; and

b. Demonstration – practical.

6. Training Aids. Engineering Watchkeeper Handouts presently located on Marine Systems Division Sharepoint site.

7. Training Limitations. Nil.

8. Test Details. Trainees will be provided feedback throughout the EO. No formal EC is included in this EO, instead the material will be tested in PC 002.

9. References. A1, A8, A13, A14, A25, B1.

10. Remarks. Nil.

11. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Lead collective operation of auxiliary systems to include:							
a.	Operating steering system remotely.			25	DP	A1 A8 A13 A14	
b.	Operating fuel oil filling/transfer system remotely to include: (1) Transferring fuel by fuel oil centrifuge; and (2) Operating the transfer pump remotely.			50			
c.	Operating fluid piping system.			25			
d.	Testing auxiliary seawater circulation automatic sequence function.			25			
e.	Testing auxiliary seawater circulation system emergency supply change over function remotely.						
TP2 Monitor ship's stability to include:							

a. Monitoring liquid level management system by: (1) Monitoring the ship's fluid systems, to include: i. Fresh water; ii. Feed water; iii. Ballast water; iv. Black and grey water; v. Diesel fuel; vi. JP5; and vii. Lube oil.	50	25	25	IL DP	A8 A13 A25 B1	
b. Assessing ship's stability by: (1) Explaining stability terminology; (2) Interpreting drafts; (3) Reading the inclinometer; (4) Describing basics of General Load Monitor (GLM); and (5) Monitoring liquid levels in damaged compartments due to flooding.		50	50			
c. Adjusting ship's stability by: (1) Transferring liquids between tanks; (2) Embarking liquids, to include: i. Fuels; ii. Fresh water; iii. Feed water; and iv. Ballast water.		50	50		A8 A13 A25 B1	
d. Recording ship's stability variables by: (1) Updating changes on GLM; (2) Recording fluid systems in the following logs: i. Ballast log; ii. Fresh water log; and iii. Fuel reports / liquid cargo log.		25				

EO 002.03 Lead Collective Operation of Ancillary Systems (DT0060)

1. Performance. Lead collective operation of ancillary systems.

2. Conditions.

   a. Given:

      (1) Operational IPMS;

      (2) Assistance;

      (3) References; and

      (4) Supervision.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) PCOT; and

      (2) MCR trainer.

3. Standard. The Engineering Watchkeeper shall independently lead collective operation of ancillary systems (DT0060) on the IPMS console, IAW refs, by:

   a. Operating ancillary systems (DT0070) to include:

      (1) Operating main seawater circulation system remotely (DT0200);

      (2) Operating main lube oil (MLO) system remotely (DT0180);

      (3) Operating the emergency functions of the MLO pumps remotely (DT0090);

      (4) Operating infrared suppression system remotely (AIT0620);

      (5) Operating gas turbine (GT) water wash system remotely (AVT0770);

      (6) Operating main propulsion GT anti-icing system remotely (QT0400);

      (7) Operating hydraulic start pack remotely (MT0020); and

      (8) Operating controllable reversible pitch propeller (CRPP) system remotely (ABT0120).

4. Time Required.

a. EO – 7 x 50 minute periods.

5. Instructional Methods.

a. Interactive lecture; and

b. Demonstration – practical.

6. Training Aids. Engineering Watchkeeper Handouts presently located on Marine Systems Division Sharepoint site.

7. Training Limitations. Nil.

8. Test Details. Trainees will be provided feedback throughout the EO. No formal EC is included in this EO, instead the material will be tested in PC 002.

9. References. A9.

10. Remarks. Nil.

11. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Operating ancillary systems to include:							
a.	Operating main seawater circulation system remotely.		25		DP	A9	Demonstrate operation of the 32mm, 150mm, and the overboard discharge valves.
b.	Operating MLO system remotely to include: (1) Flashing pumps; and (2) Managing tank levels.		25	25	DP		
c.	Operating emergency functions of the MLO pumps remotely to include: (1) Air turbine test; and (2) Standby pump test.	10	25	25	IL DP		
d.	Operating infrared suppression system remotely.	10	10	10	IL DP		
e.	Operating GT water wash system remotely.	15	25	25	IL DP		

f.	Operating GT anti-icing system remotely.	10	10	10	IL DP		
g.	Operating HSS remotely.		15	25	DP		
h.	Operating CRPP system remotely by: (1) Explaining normal oil flow; (2) Describing the following pumps: i. Main pumps; ii. Standby pumps; iii. Header pump; and iv. Hand pump. (3) Describing modes of operation to include: i. Normal operation; and ii. Emergency operation.	50			IL		

EO 002.04 Operate Power Generation & Distribution (PG&D) System

1. Performance. Operate PG&D system.

2. Conditions.

   a. Given:

      (1) Operational IPMS;

      (2) Assistance;

      (3) References; and

      (4) Supervision.

   b. Denied: Nil.

   c. Environmental:

      (1) PCOT; and

      (2) MCR Trainer.

3. Standard. The Engineering Watchkeeper shall independently operate the PG&D system, IAW refs by:

   a. Operating generators in all modes (CT0280); and

   b. Configuring ship's electrical load (CT0060).

4. Time Required.

   a. EO – 1 x 50 minute period.

5. Instructional Methods. Demonstration – practical.

6. Training Aids. Engineering Watchkeeper Handouts presently located on Marine Systems Division Sharepoint site.

7. Training Limitations. Nil.

8. Test Details. Trainees will be provided feedback throughout the EO. No formal EC is included in this EO, instead the material will be tested in PC 002.

9. References. A4, A13, A15, A21, A22, A23, A40, A41, A42, A44.

10. Remarks. Nil.

11. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Operate generators in all modes to include:							
a. Operating generators in all modes by: (1) Selecting automatic mode; (2) Selecting manual mode; (3) Selecting manual permissive mode; and (4) Selecting emergency operation mode.			10	15	DP	A4 A13 A15 A21 A22 A23 A40 A41 A42 A44	
TP2 Configure ship's electrical load to include:							
a. Configuring ship's electrical load by: (1) Switching the power supplies between two Diesel Generators (DG's) in: i. Automatic; ii. Manual; iii. Manual permissive; and iv. Emergency operation.			10	15	DP	A4 A13 A15 A21 A22 A23 A40 A41 A42 A44	

EO 002.05 Operate Gearing & Shafting Systems

1. Performance. Operate gearing & shafting systems.

2. Conditions.

   a. Given:

      (1) Operational IPMS;

      (2) Assistance;

      (3) References; and

      (4) Supervision.

   b. Denied: Nil.

   c. Environmental:

      (1) PCOT; and

      (2) MCR Trainer.

3. Standard. The Engineering Watchkeeper shall independently operate the gearing and shafting system, IAW refs by:

   a. Leading operation of gearing and shafting systems (ABT0050) to include:

      (1) Operating main gearing and shafting systems (ABT0170);

      (2) Testing operation of gearbox (BT0200);

      (3) Operating gearbox clutch remotely (ABT0150);

      (4) Monitoring shaft line operation (ABT0080); and

      (5) Monitoring gearbox operation remotely (ABT0070).

4. Time Required.

   a. EO – 5 x 50 minute periods.

5. Instructional Methods.

   a. Interactive lecture; and

   b. Demonstration – practical.

6. Training Aids. Engineering Watchkeeper Handouts presently located on Marine Systems Division Sharepoint site.

7. Training Limitations. Nil.

8. Test Details. Trainees will be provided feedback throughout the EO. No formal EC is included in this EO, instead the material will be tested in PC 002.

9. References. A2, A7, A17, A18, A21, A30.

10. Remarks. Nil.

11. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Lead operation of gearing and shafting systems to include:							
a.	Operating main gearing and shafting systems by: (1) Monitoring towing operations; (2) Monitoring single shaft operations; and (3) Steering by main engines.		50		DP	A2 A7 A17 A18 A21 A30	Instructor to run trainees through all applicable IPMS pages and system operating parameters.
b.	Testing operation of gearbox by: (1) Testing functionality of the Power Turbine (PT) brake remotely; and (2) Conducting auto and manual centralization.		50		DP	A2 A7 A17 A18 A21 A30	Instructor to demonstrate remote operations of gearbox functions.
c.	Operating gearbox clutch remotely by: (1) Operating friction clutches remotely; and (2) Operating Synchronous Self Shifting (SSS) locking oil valves remotely.		50		DP	A2 A7 A17 A18 A21 A30	Instructor to demonstrate remote operations of clutches.
d.	Monitoring shaft line operation.		50		DP	A2 A17 A18 A21	Instructor to run trainees through all applicable IPMS pages

					and system operating parameters.
e. Monitoring gearbox operation remotely.		50	DP	A2 A7 A17 A18 A21 A30	Instructor to run trainees through all applicable IPMS pages and system operating parameters.

EO 002.06 Operate Main Propulsion System

1. Performance. Operate main propulsion system.

2. Conditions.

   a. Given:

      (1) Operational IPMS;

      (2) Operational Local Operating Panel (LOP);

      (3) Assistance;

      (4) References; and

      (5) Supervision.

   b. Denied: Nil.

   c. Environmental:

      (1) PCOT; and

      (2) MCR Trainer.

3. Standard. The Marine Technician Watchkeeper shall operate the main propulsion system independently, IAW refs by:

   a. Operating main propulsion system in all modes (RT0010) by:

      (1) Operating propulsion control remotely (ABT0210) to include:

         i. Operating main propulsion GT remotely (QT0420); and

         ii. Operating main diesel engine remotely (QT0390).

      (2) Operating propulsion control locally (ABT0200) to include:

         i. Operating main propulsion GT locally (QT0410); and

         ii. Operating main diesel engine locally (QT0350).

4. Time Required.

   a. EO – 13 x 50 minute periods.

5. Instructional Methods.

a. Interactive lecture;

b. Role playing; and

c. Demonstration – practical.

6. Training Aids. Engineering Watchkeeper Handouts presently located on Marine Systems Division Sharepoint site.

7. Training Limitations. Nil.

8. Test Details. Trainees will be provided feedback throughout the EO. No formal EC is included in this EO, instead the material will be tested in PC 002.

9. References. A2, A9, A16, A23, A36.

10. Remarks. Nil

11. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Operate the main propulsion system in all modes to include:							
a. Operating the main propulsion system remotely by: (1) Operating the main propulsion GT remotely to include; and i. Engine prestart requirements; ii. Automatic engine start procedures; iii. Automatic drive mode changeover; iv. Single engine operation; v. Two engine operation; and vi. Automatic engine shut down procedures. vii. Semi-automatic operation (2) Operating the main diesel engine remotely to include:		20 0	25	100	IL DP RP	A2 A9 A16 A23 A36	Instruction to include all operating parameters, start permissives, engine trips, and restrictions. Instructor to demonstrate all drive mode configurations.

i. Engine prestart requirements; ii. Automatic engine start procedures; iii. Automatic drive mode changeover; and iv. Automatic engine shut down procedure. v. Semi-automatic operation						
b. Operating the main propulsion system locally by: (1) Operating the main propulsion GT locally to include; and i. Engine prestart requirements; ii. Manual engine start procedures; iii. Manual drive mode changeover. (2) Operating the main Diesel Engine locally to include: i. Engine prestart requirements; ii. Manual engine start procedures; and iii. Manual drive mode changeover.	50	25	250	IL DP RP	A2 A9 A16 A23 A36	Instruction to include all operating parameters, start permissives, engine trips and restrictions. Instructor to demonstrate starting of engines from LOPs and manual drive mode changeovers locally from LOPs.

PERFORMANCE CHECK – PC 002

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitation Instructions

1. PCOT with 12 training computers.

2. Two invigilators.

Arrangements and Procedures

3. PC 002 will be a practical assessment based on content from PO 002.

4. Trainees required to attain 76 points.

5. Time required: 2 x 50 minute periods (1 trainee per computer with two invigilators overlooking the assessment of all trainees).

6. References will be available.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. PC 002 will be a practical assessment based on content from PO 002.

2. During this practical assessment, you will be required to operate IPMS console. You must operate IPMS console in accordance with the standard.

3. References will be provided.

4. Trainees required to attain 76 points.

5. You will have 100 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Give guidance for the development of the assessment.

1. See instructions to the evaluator located in the table below.

PART IV – PERFORMANCE CHECK LIST

Check List PC 002	SAT	UN-SAT	Comments (for invigilator, before printing this check list these comments to be deleted and this space used for invigilator comments).
Flash-up the ship from 4hrs NFP using the trainer and bring to immediate notice for power by:			
Conducting pre-sail checks with the IPMS pre-sail check list.			The trainee is to use the check off list provided to go through all systems and prepare the ship for flash-up.
Conducting **SAFETY** brief with the flash-up watch.			This is a critical check off, if the trainee does not provide the proper brief it is considered a failure of this EC. Brief to be clear to all involved personnel.
Navigating through pages.			Does the trainee know how to effectively use the IPMS to navigate through all the pages?
Changing station in control.			Have the trainee pass and offer station in control twice.
Flashing-up auxiliary equipment.			The trainee is to flash-up or instruct the roundsperson to flash-up auxiliary equipment in support of flashing up the plant to immediate notice for power.
Flashing-up ancillary equipment.			The trainee is to flash-up ancillary equipment to include trialing the MLO emergency function and verifying that the maintenance watch conducted their flash-up procedure to prove the ancillary equipment.
Preparing main engines for flash-up.			The trainee to prepare the main engines for flash-up.
Instructing the roundsperson to bar over the PDE.			Instruct the roundsperson to bar over the PDE.
Bringing the PDE to idle.			Bring the PDE to idle.
Dry motoring the Gas Turbines.			Dry motor the Gas Turbines.
Flashing the gas turbines to idle.			The trainee is to flash-up the gas turbines to idle. Once the engines are at idle the friction clutches are to be centralized. Once all conditions are met for immediate notice for power, the trainee is to inform the instructor.
Instructions to Evaluator: **1.** The instructor is to have the trainee flash-up the ship using the trainer and bring to immediate notice for power; **2.** The trainee shall meet each requirement in the checklist. If the trainee fails to provide a proper safety brief to the flash-up watch, it will be considered a fail; **3.** Trainees begin with 100 points; each fault subtracts 7 from the total; and **4.** The trainee must receive more than 75 points in total to pass this PC.			

The only grade awarded for this PC is PASS (for totals 76 points or greater) or FAIL (for totals less than 76 points).

PO 003 INVESTIGATE ENGINEERING SYSTEMS

EO 003.01 Investigate collective operation of auxiliary systems (CT0130)

1. Performance. Investigate collective operation of auxiliary systems.

2. Conditions.

   a. Given:

      (1) References;

      (2) Assistance;

      (3) Operational IPMS; and

      (4) Supervision.

   b. Denied: Nil.

   c. Environmental:

      (1) PCOT; and

      (2) MCR trainer.

3. Standard. The Engineering Watchkeeper shall independently investigate engineering systems, IAW refs by:

   a. Troubleshooting auxiliary diesel engine internal components (AVT1330), to include:

      (1) Troubleshooting auxiliary diesel engine control system (AVT1290);

      (2) Troubleshooting auxiliary diesel engine sea water cooling system (AVT1360);

      (3) Troubleshooting auxiliary diesel engine exhaust system (AVT1300);

      (4) Troubleshooting auxiliary diesel engine starting air system (AVT1350);

      (5) Troubleshooting auxiliary diesel engine fresh water system (AVT1320);

      (6) Troubleshooting auxiliary diesel engine fuel system (AVT1310);

      (7) Troubleshooting auxiliary diesel engine lube oil system (AVT1340); and

      (8) Troubleshooting auxiliary diesel engine combustion air system (AVT1280).

   b. Troubleshooting auxiliary seawater circulation system (AHT0830);

c. Troubleshooting seawater service system (firemain) (AHT0890); and

d. Troubleshooting bilge system (AHT0840).

4. Time Required. EO – 9 x 50 minute periods.

5. Instructional Methods.

a. Interactive lecture;

b. Role playing; and

c. Demonstration – practical.

6. Training Aids. Engineering Watchkeeper Handouts presently located on Marine Systems Division Sharepoint site.

7. Training Limitations. Nil.

8. Test Details. EO 003.01 will be assessed in PC 004

9. References. A1, A26, A40, A41, A42

10. Remarks. Nil.

11. Teaching Points

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Troubleshoot auxiliary diesel engine internal components to include:							
a. Rectifying diesel engine alarms by: (1) Troubleshooting auxiliary diesel engine control system; (2) Troubleshooting auxiliary diesel engine sea water cooling system; (3) Troubleshooting auxiliary diesel engine exhaust system; (4) Troubleshooting auxiliary diesel engine starting air system; (5) Troubleshooting auxiliary diesel engine fresh water system;		20 0		100	IL DP RP	A40 A41	Instructor to assume role as command for student briefing in practical phase.

(6) Troubleshooting auxiliary diesel engine fuel system; (7) Troubleshooting auxiliary diesel engine lube oil system; and (8) Troubleshooting auxiliary diesel engine combustion air system.					
TP2 Troubleshoot auxiliary seawater circulation system to include:					
a. Assessing reasons for possible system pressure loss to include: (1) Pump failure; (2) Pipe rupture; and (3) Clogged strainer. b. Determining Effects on other machinery.	25	25	IL DP RP	A1	Instructor to assume role as command for student briefing in practical phase.
TP3 Troubleshoot seawater service system to include:					
a. Assessing reasons for possible system pressure loss to include: (1) Pump failure; (2) Clogged strainer; (3) Firemain rupture; and (4) Splitting firemain system.	25	25	IL DP RP	A42	Instructor to assume role as command for student briefing in practical phase.
TP4 Troubleshoot bilge system to include:					
a. Managing the oil pollution and abatement system to include consolidating bilges.	25	25	IL DP	A26	

EO 003.02 Investigate collective operation of ancillary systems (CT0120)

1. Performance. Investigate collective operation of ancillary systems.

2. Conditions.

   a. Given:

      (1) References;

      (2) Assistance;

      (3) Operational IPMS; and

      (4) Supervision.

   b. Denied: Nil.

   c. Environmental:

      (1) PCOT; and

      (2) MCR Trainer.

3. Standard. The Engineering Watchkeeper shall independently investigate engineering systems, IAW refs by:

   a. Troubleshooting ancillary systems (AHT0820) to include:

      (1) Troubleshooting GT water wash system (AVT1430);

         i. Verifying operating parameters of GT water wash system (AVT1450);

      (2) Troubleshooting main seawater circulation system (AHT0880);

      (3) Troubleshooting main lube oil system (MLO) (AHT0870); and

      (4) Troubleshooting MLO filling and transfer system (AHT0850).

4. Time Required. EO – 6 x 50 minute periods.

5. Instructional Methods.

   a. Interactive lecture;

   b. Role playing; and

   c. Demonstration – practical.

6. Training Aids. Engineering Watchkeeper Handouts presently located on Marine Systems Division Sharepoint site.

7. Training Limitations. Nil.

8. Test Details. EO 003.02 will be assessed in PC 004.

9. References. A3, A9, A11, A12, A31

10. Remarks. Nil.

11. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Troubleshoot ancillary systems to include:							
a.	Troubleshooting GT water wash system by: (1) Adjusting pressure regulator; (2) Verifying operating parameters of the GT water wash system; and (3) Switching HSS to manual.	25	25		IL DP	A3 A9 A11 A12	
b.	Troubleshooting main seawater circulation system to include: (1) Monitoring MLO temperature; (2) Identifying pump failure; (3) Identifying pipe rupture; and (4) Identifying clogged strainer.	25	25		IL DP	A9 A31	
c.	Troubleshooting MLO system by: (1) Troubleshooting MLO fill and transfer to include: i. Managing tank levels; ii. Transferring MLO; and iii. Investigating centrifuge group warnings. (2) Investigating: i. Loss of MLO pressure; ii. Loss of GT synthetic lube oil; and iii. High bearing temperature.	100	100		IL DP RP	A9 A31	Instructor to assume role as command for student briefing in practical phase.

EO 003.03 Investigate collective operation of power generation and distribution system

1. Performance. Investigate collective operation of PG&D system.

2. Conditions.

   a. Given:

      (1) References;

      (2) Assistance;

      (3) Operational IPMS; and

      (4) Supervision.

   b. Denied: Nil.

   c. Environmental:

      (1) PCOT; and

      (2) MCR Trainer.

3. Standard. The Engineering Watchkeeper shall independently investigate engineering systems, IAW refs by:

   a. Investigating collective operation of power generation and distribution system (CT0150) to include:

      (1) Investigating switchboard auto functions; and

      (2) Configuring ship's electrical load.

4. Time Required. EO – 1.5 x 50 minute periods.

5. Instructional Methods.

   a. Interactive lecture; and

   b. Demonstration – practical.

6. Training Aids. Engineering Watchkeeper Handouts presently located on Marine Systems Division Sharepoint site.

7. Training Limitations. Nil.

8. Test Details. EO 003.03 will be assessed in PC 004.

9. References. A4, A13, A15, A21, A22, A23, A40, A41, A42, A44.

10. Remarks. Nil

11. Teaching Points

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Investigate engineering systems to include:							
a.	Investigating collective operation of power generation and distribution system (CT0150) by: (1) Investigating switchboard auto functions; (2) Investigating electrical pages to include: and i. PG&D; ii. ACOS; iii. MCOS; iv. UPS; and v. Static Frequency Converter (SFC). (3) Describing effects on other systems (can include CSE equipment).	50	25		IL DP	A4 A13 A15 A21 A22 A23 A40 A41 A42 A44	

EO 003.04 Investigate operation of gearing and shafting systems (CT0160)

1. Performance. Investigate operation of gearing and shafting systems.

2. Conditions.

   a. Given:

      (1) References;

      (2) Assistance;

      (3) Operational IPMS; and

      (4) Supervision.

   b. Denied: Nil.

   c. Environmental:

      (1) PCOT; and

      (2) MCR Trainer.

3. Standard. The Engineering Watchkeeper shall independently investigate operation of gearing and shafting systems, IAW refs by:

   a. Evaluating gearbox function (BT0050); and

   b. Evaluating shaftline function (BT0060).

4. Time Required. EO – 3 x 50 minute periods.

5. Instructional Methods.

   a. Interactive lecture; and

   b. Demonstration – practical.

6. Training Aids. Engineering Watchkeeper Handouts presently located on Marine Systems Division Sharepoint site.

7. Training Limitations. Nil.

8. Test Details. EO 003.04 will be assessed in PC 004.

9. References. A2, A7, A9, A18, A19, A20, A23.

10. Remarks. Nil

11. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Evaluate gearbox function to include:							
a. Troubleshooting bearing temperatures by: (1) Investigating bearing alarm and warning values.			50		DP	A2 A7 A9 A23	Instructor to take the trainees through bearing alarm and warning values and what to look for to evaluate whether a bearing has possibly wiped or not.
TP2 Evaluate shaftline function to include:							
a. Troubleshooting plummer block temperatures by: (1) Investigating bearing warnings and alarms.			50		DP	A2 A9 A18 A19 A20 A23	Instructor to take the trainees through bearing alarm and warning values.
b. Troubleshooting shaftline torque by: (1) Investigating torque readings.			50		DP	A2 A9 A18 A19 A20 A23	Instructor to take the trainees through the warning and alarm values of shaft torque and what might cause torque to change.

EO 003.05 Investigate Main Propulsion Systems

1. Performance. Investigate main propulsion systems.

2. Conditions.

   a. Given:

      (1) Operational IPMS trainer;

      (2) Assistance;

      (3) References; and

      (4) Supervision.

   b. Denied: Nil.

   c. Environmental:

      (1) PCOT; and

      (2) MCR Trainer.

3. Standard. The Engineering Watchkeeper shall independently investigate main propulsion systems, IAW refs by:

   a. Investigating collective operation of main propulsion systems (CT0140) to include:

      (1) Troubleshooting main propulsion system (CT0430);

      (2) Troubleshooting fuel oil system (AHT0860);

      (3) Troubleshooting propulsion diesel engine fire detection control system (ALT2180);

      (4) Troubleshooting propulsion diesel engine enclosure carbon monoxide sensor (ALT2170); and

      (5) Troubleshooting propulsion diesel engine low pressure control air (ALT2200).

4. Time Required. EO – 7 x 50 minute periods.

5. Instructional Methods.

   a. Interactive lecture; and

   b. Demonstration – practical.

6. Training Aids. Engineering Watchkeeper Handouts presently located on Marine Systems Division Sharepoint site.

7. Training Limitations. Nil.

8. Test Details. EO 003.05 will be assessed in PC 004.

9. References. A9, A16, A31.

10. Remarks. Nil

11. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Investigate collective operation of main propulsion systems to include:							
a.	Troubleshooting main propulsion system by: (1) Explaining engineering problems to include: i. Lube oil failure; ii. Fuel leak; iii. Engine over speed condition; iv. High PTIT; v. High bearing temperature; vi. High vibration; vii. Hot start; viii. High air intake differential pressure; and ix. PDE high fresh water temperature. (2) Troubleshooting propulsion diesel engine fire detection control system; (3) Troubleshooting propulsion diesel engine enclosure carbon monoxide sensor; and (4) Troubleshooting propulsion diesel engine low pressure control air.	100	200		IL DP	A9 A13 A31	

b. Troubleshooting fuel oil system to include: (1) Explaining loss of main fuel system pressure to include; i. Loss of fuel pump; ii. Emergency fuel tank auto sequence; and iii. Utilizing transfer pump.	25	25	IL DP		

PERFORMANCE CHECK – PO 003

The concepts in PO 003 will be tested in PC 004.

PO 004 RESPOND TO ALARMS AND EMERGENCIES

EO 004.01 Respond to Auxiliary and Ancillary System Alarms

1. Performance. Respond to auxiliary and ancillary system alarms.

2. Conditions.

   a. Given:

      (1) Operational IPMS;

      (2) References;

      (3) Assistance; and

      (4) Supervision.

   b. Denied: Nil.

   c. Environmental:

      (1) PCOT; and

      (2) MCR trainer.

3. Standard. The Engineering Watchkeeper shall respond to auxiliary and ancillary system alarms independently, IAW refs by:

   a. Responding to minor alarms and faults;

   b. Responding to auxiliary and ancillary emergencies IAW with "Manual of Emergency Procedures, Marine Systems Engineering (at sea for Halifax class ships)"; and

   c. Advising command on auxiliary and ancillary systems by:

      (1) Advising command on HVAC/CBRN air distribution system capabilities and limitations (JT0040);

      (2) Advising command on chilled water and hydronic system capabilities and limitations (JT0020); and

      (3) Advising command on hotel system capabilities and limitations (JT0030).

4. Time Required.

   a. EO – 12.5 x 50 minute periods; and

   b. EC – 50 minutes per trainee.

5. Instructional Methods.

a. Interactive lecture;

b. Demonstration – practical; and

c. Role playing.

6. Training Aids.

a. A physical copy of Manual of Emergency Procedures Marine Systems Engineering (at sea) C-03-050-HFX/MB-002.

7. Training Limitations. Nil.

8. Test Details. EO 004.01 will be tested in EC 004.01.

9. References. A1, A3, A4, A8, A13, A14, A24, A25, A26, A27, A28, A29, A31, A42.

10. Remarks. Nil

11. Teaching Points

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Respond to minor alarms and faults to include:							
a. Responding to auxiliary and ancillary alarms and faults for the following: (1) Sea water system (Firemain); (2) Fresh water system; (3) Black water system; (4) Auxiliary seawater circulation system; (5) Hydronic/chilled water system; (6) Refrigeration system; (7) Main lube oil; (8) Fuel oil service system; (9) Fuel filter water separator; (10) Oil pollution abatement system; (11) Service steam system; (12) HP air system; (13) LP air system; (14) HVAC/CBRN system; and			50	100	DP	A1 A3 A4 A8 A13 A14 A24 A25 A26 A27 A28 A29 A42	

(15) Damage control firefighting systems.						

TP2 Respond to auxiliary and ancillary emergencies by:						
a. Responding to the following auxiliary and ancillary emergencies IAW manual of emergency procedures, MSE(at sea for Halifax class ships): (1) MLO failure; (2) Loss of main fuel system pressure; (3) Loss of HPA; and (4) Loss of LPA.	50	100	100	IL DP	A31	Trainees to receive a physical copy of Manual of Emergency Procedures – MSE (at sea) as a reference tool.
TP3 Advise command on auxiliary and ancillary systems by:						
a. Advising command on HVAC/ CBRN air distribution system capabilities and limitations (JT0040), to include the following topics: (1) Current system configuration; (2) Current system deficiencies; and (3) Possible solutions or work arounds to repair current deficiencies.		25	50	DP RP	A29 A31	Instructor to assume role as command for student briefing in practical phase.
b. Advising command on chilled water and hydronic system capabilities and limitations (JT0020), to include the following topics: (1) Current system configuration; (2) Current system deficiencies; and (3) Possible solutions or work arounds to repair current deficiencies.		25	50	DP RP	A24 A31	Instructor to assume role as command for student briefing in practical phase.
c. Advising command on hotel system capabilities and limitations (JT0030), to include the following topics: (1) Fresh water system; (2) Service steam system; and (3) Black water system.		25	50		A25 A27 A28 A31	

Enabling Check – EC 004.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitation Instructions

1. MCR Trainer.

Arrangements and Procedures

2. EC 004.01 will be a practical assessment based on content from EO 004.01.

3. The trainee must achieve 76 points.

4. Time required: 50 minutes per trainee.

5. References will be available.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 004.01 will be a practical assessment based on content from EO 003.01, 003.02 and 004.01.

2. During this practical assessment, you will be required to respond to auxiliary and ancillary alarms. You must pass this test to pass EO 004.01. To pass the test, you must complete response to auxiliary and ancillary alarms in accordance with the standard.

3. References will be available.

4. The trainee must achieve 76 points.

5. IAW EC checklist

6. You will have 50 minutes to complete this assessment.

7. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

See instructions to the evaluator located in the table below.

PART IV – ENABLING CHECK LIST

Check List	SAT	UNSAT	Comments
Review engineering log.			The trainee is to review the engineering log and update as necessary.
Perform page rounds.			The trainee is to navigate IPMS pages and report any deficiencies. Trainee to take appropriate SIC and role attribution.
Effective Communication.			The trainee is to communicate to the MCR, personnel, and OOW clearly and concisely.
Rectify diesel engine alarms.			Ensure diesel fault is added.
Respond to auxiliary and ancillary alarms and faults.			The trainee is to demonstrate understanding of the scenario cause and effect.
Respond to auxiliary and ancillary emergencies (**SAFETY**).			
Perform GT water wash.			
Advise command on HVAC/ CBRN air distribution system capabilities and limitations.			
Advise command on chilled water and hydronic system capabilities and limitations.			
Advise command on hotel system capabilities and limitations.			

Instructions to Evaluator:

1. The instructor is to have the trainee in the MCR trainer and respond to various procedures, briefs, faults, and alarms, including one emergency drill and three minor alarms;
2. The fault can be anything taught on the EO. Students are to receive one diesel fault and two other faults;
3. Trainees begin with 100 points; each ‘UNSAT’ subtracts 7 from the total;
4. The trainee must receive more than 75 points in total to pass this EC; and
5. The only grades awarded for this EC are PASS (for totals 76 points or greater) or FAIL (for totals less than 76 points).

EO 004.02 Respond to PG&D alarms

1. Performance. Respond to PG&D alarms.

2. Conditions.

   a. Given:

      (1) References;

      (2) Supervision;

      (3) Assistance; and

      (4) Operational IPMS.

   b. Denied: Nil.

   c. Environmental:

      (1) PCOT; and

      (2) MCR Trainer.

3. Standard. The trainee shall respond to PG&D alarms independently, IAW refs by:

   a. Advising command on PG&D systems (CT0030) to include:

      (1) Advising command on ship's power configuration (PT0010);

      (2) Advising command on ship's electrical load (CT0050); and

      (3) Reporting on PG&D system (CT0330).

4. Time Required.

   a. EO – 4 x 50 minute periods; and

   b. EC – 50 minutes per trainee.

5. Instructional Methods.

   a. Role playing; and

   b. Demonstration – practical.

6. Training Aids.

a. A physical copy of Manual of Emergency Procedures Marine Systems Engineering (at sea) C-03-050-HFX/MB-002.

7. Training Limitations. Nil.

8. Test Details. EO 004.02 will be tested in EC 004.02.

9. References. A15, A31, A46.

10. Remarks. Nil.

11. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Respond to PG&D alarms to include:							
a. Advising command on PG&D systems (CT0030) by: (1) Advising command on ship's power configuration (PT0010); (2) Advising command on ship's electrical load (CT0050); and (3) Reporting on PG&D system (CT0330). b. Responding to emergencies: (1) High crankcase pressure (2) Enclosure fire (include FWS and new NOVEC system)			25	50	DP RP	A15 A31 A46	Instructor to assume role as command for student briefing in practical phase.
TP2 Respond to total power failure as an Engineering Watchkeeper by:							
a. Verifying total blackout conditions; b. Verifying auto sequences to include: (1) Generators auto sequence; (2) MLO auto sequence; (3) FOS auto sequence; (4) CRPP standby pumps; and (5) Diesel driven fire pumps (DDFP). c. Verifying load shed breakers are reset; and d. Restoring plant configuration.			25	100	DP RP	A15 A31 A46	Instructor to assume role as command for student briefing in practical phase.

Enabling Check – EC 004.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitation Instructions

1. MCR Trainer.

Arrangements and Procedures

2. EC 004.02 will be a practical assessment based on content from EO 003.03 and 004.02.

3. Pass/Fail.

4. Time required: 50 minutes.

5. References will be available.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 004.02 will be a practical assessment based on content from EO 003.03 and 004.02.

2. During this practical assessment, you will be required to respond to PG&D alarms. You must pass this test to pass EO 004.02. To pass the test, you must respond to PG&D alarms in accordance with the standard.

3. References will be provided.

4. Pass/Fail IAW EC checklist.

5. You will have 50 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

See instructions to the evaluator located in the table below.

PART IV – ENABLING CHECK LIST

Check List	SAT	UNSAT	Comments
Review engineering log.			The trainee is to review the engineering log and update as necessary.

Perform page rounds.			The trainee is to navigate IPMS pages and report any deficiencies.
Effective Communication.			The trainee is to communicate to the MCR, personnel, and OOW clearly and concisely.
Respond to PG&D alarms and faults.			The trainee is to demonstrate understanding of the scenario cause and effect.
Respond to blackout			
Advise command on PG&D system capabilities and limitations.			

Instructions to Evaluator:

1. The instructor is to have the trainee in the MCR trainer and respond to various procedures, briefs, faults, and alarms, including one blackout;
2. The fault can be anything taught on the EO;
3. Trainees begin with 100 points; each ‘UNSAT’ subtracts 7 from the total;
4. The trainee must receive more than 75 points in total to pass this EC; and
5. The only grades awarded for this EC are PASS (for totals 76 points or greater) or FAIL (for totals less than 76 points).

EO 004.03 Respond to Main Propulsion Alarms

1. Performance. Respond to main propulsion alarms.

2. Conditions.

   a. Given:

      (1) References;

      (2) Supervision;

      (3) Assistance; and

      (4) Operational IPMS.

   b. Denied: Nil.

   c. Environmental:

      (1) PCOT; and

      (2) MCR Trainer.

3. Standard. The Engineering Watchkeeper shall respond to main propulsion alarms independently, IAW refs by:

   a. Advising command on machinery/system status (GT0010) to include: and

      (1) Advising command on propulsion capabilities and limitations (CT0040);

      (2) Advising command on marine systems operational capabilities and limitations (CT0020); and

      (3) Advising command on marine systems engineering parameters (CT0010).

   b. Reporting on propulsion deficiencies (CT0340) to include:

      (1) Reporting on machinery limitations (CT0310); and

      (2) Reporting on marine systems engineering parameters (CT0320).

4. Time Required.

   a. EO – 6 x 50 minute periods; and

   b. EC – 150 minutes per trainee.

5. Instructional Methods.

a. Interactive lecture;

b. Role playing; and

c. Demonstration – practical.

6. Training Aids. A physical copy of Manual of Emergency Procedures Marine Systems Engineering (at sea) C-03-050-HFX/MB-002.

7. Training Limitations. Nil.

8. Test Details. EO 004.03 will be assessed in EC 004.04.

9. References. A2, A7, A9, A31.

10. Remarks. Nil.

11. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Advise command on machinery/system status to include:							
a. Advising command on propulsion capabilities and limitations by: (1) Adhering to drive modes limitations; (2) Impact on overall plant; (3) Explaining optimum engine speeds; and (4) Advising command on marine systems engineering parameters.		50	25	100	IL DP RP	A9 A31	Instructor to assume role as command for student briefing in practical phase.
b. Advising command on marine systems operational capabilities and limitations to include explaining impact of failures or emergencies to the operation of the propulsion plant.		25	25	50	IL DP RP	A9 A31	
TP2 Report on propulsion deficiencies to include:							
Reporting on machinery limitations by reporting on marine systems engineering parameters; given the altered state of the plant due to emergencies/problems, what it means to the current capability.			25		DP RP	A9	Instructor to assume role as command for student briefing in practical phase.

EO 004.04 Respond to Shipboard Emergencies

1. Performance. Respond to shipboard emergencies

2. Conditions.

   a. Given:

      (1) Operational IPMS;
      (2) References;
      (3) Assistance; and
      (4) Supervision.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) PCOT; and
      (2) MCR Trainer.

3. Standard. The Engineering Watchkeeper shall respond to shipboard emergencies independently, IAW refs by:

   a. Managing propulsion plant operation in action and emergency stations (IT0470); and

   b. Advising command on ship's stability (WT0020).

4. Time Required.

   a. EO – 3 x 50 minute periods; and

   b. EC – 150 minutes per trainee.

5. Instructional Methods.

   a. Interactive lecture;

   b. Role playing; and

   c. Demonstration – practical.

6. Training Aids.

   a. A physical copy of Manual of Emergency Procedures Marine Systems Engineering (at sea) C-03-050-HFX/MB-002; and

b. Engineering Watchkeeper aide de memoire.

7. Training Limitations. Nil.

8. Test Details. EO 004.04 will be tested in EC 004.04.

9. References. A10, A13, 25, A26, A30, A46, A47, B1.

10. Remarks. Nil.

11. Teaching Points:

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Manage propulsion plant operation in action and emergency stations to include:							
a. Responding to the following ship's emergencies: (1) Action stations; (2) Emergency stations; (3) Emergency flying stations; (4) Fuel/HAZMAT spill; and (5) Rescue stations.		25	25		IL DP	A46 A47	
TP2 Advise command on ship's stability to include:							
a. Recommending the following stability conditions by: (1) Transferring liquid loads; (2) Changing the ships course; and (3) Commencing pumping operations.		25	25	50	IL DP RP	A10 A13 A25 A26 A30 B1	Instructor to assume role as command for student briefing in practical phase.

Enabling Check – EC 004.04

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitation Instructions

1. MCR trainer.

Arrangements and Procedures

2. EC 004.04 will be a practical assessment based on content from EO 003.04, 003.05, 004.03 and 004.04.

3. The trainee must achieve 76 points.

4. Time required: 150 minutes per student.

5. References will be available.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. EC 004.04 will be a practical assessment based on content from EO 004.03 and 004.04.

2. During this practical assessment, you will be required to respond to main propulsion alarms and ship emergencies. You must pass this test to pass EO 004.03 and 004.04. To pass the test, you must respond to main propulsion alarms and ship emergencies in accordance with the standard.

3. References will be provided.

4. The trainee must achieve 76 points. IAW EC checklist

5. You will have 150 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

See instructions to the evaluator located in the table below.

PART IV – ENABLING CHECK LIST

Check List	SAT	UNSAT	Comments
Review engineering log.			The trainee is to review the engineering log and update as necessary.
Perform page rounds.			The trainee is to navigate IPMS pages and report any deficiencies.
Effective communication.			The trainee is to communicate with the MCR personnel and OOW in a clear and concise manner, showing confidence.
Manage propulsion plant operation in action and emergency stations.			The trainee is to select the appropriate drive mode for the situation: - Action stations; - Emergency stations; - Rescue stations; - Flying stations; and - Fuel/HAZMAT spill.
Advise command on propulsion capabilities and limitations.			The trainee is to explain to the OOW drive mode and speed capabilities/limitations.
Advise command on marine systems operational capabilities and limitations.			The trainee is to explain operational impact of the following situations that can be selected: - MLO failure; - Fuel leak; - Engine over speed; - High PTIT; - High bearing temperature; - High vibration; - Hot start; - High air intake differential pressure; - Propulsion engine enclosure fire; - GT internal fire; - CRPP failure; and - PDE high fresh water temperature.
Report on propulsion deficiencies.			Situation – Impact – Recommendation – Limitations standard of reporting
Report on machinery limitations.			Situation – Impact – Recommendation – Limitations standard of reporting
Advise command on ship's stability.			The trainee is to assess the ships current list and bring up the tanks overview page.

Instructions to Evaluator:

1. The instructor is to have the trainee in the MCR trainer and respond to various procedures, briefs, faults, and alarms to include three drills;
2. The scenarios selected should challenge the trainees knowledge and understanding of the system; as long as it was taught on the EO;
3. Trainees begin with 100 points; each fault subtracts 7 from the total;
4. The trainee must receive more than 75 points in total to pass this EC; and
5. The only grades awarded for this EC are PASS (for totals 76 points or greater) or FAIL (for totals less than 76 points).

PERFORMANCE CHECK – PC 004

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Operational Ship.

Arrangements and Procedures

2. PC 004 will be a practical on-board assessment based on content from PO 001, PO 002, PO 003 and PO 004.

3. Pass/Fail.

4. Time required: six months at sea on an operational HFX Class ship.

5. References will be provided.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

PRACTICAL ASSESSMENT

1. PC 004 will be a practical assessment based on content from PO 001, PO 002, PO 003 and PO 004.

2. During this practical assessment, you will be required to operate IPMS console. You must perform watchkeeping duties in accordance with the standard.

3. References will be provided.

4. Pass/Fail.

5. You will have six months at sea to complete this assessment.

6. This assessment will be completed upon being posted to an operational HFX Class ship.

PART III – PERFORMANCE CHECK NOTES

The following checklist is a copy of Appendix 1 of Annex E which includes additional notes regarding assessment criteria. This checklist is to be used by the invigilator (usually the MSE Training PO) on-board ship to help with the assessment of the trainee.

PO/EO	Performance Statement	Comments
002.01	The Trainee shall Conduct Watchkeeping operations to the satisfaction of the MSEO by::	
002.01.001	Conduct pre-watch and on-watch rounds.	Training PO to conduct a set of rounds with the Trainee and be satisfied with the Trainee's knowledge
002.01.002	Conduct flash-up and shut-down routine.	Trainee to conduct pre-sail flash-up of the plant as Engineering Watchkeeper as well as shut down the plant at the end of a sail.
002.01.003	Operate propulsion system in automatic control mode.	Trainee to prove their ability to stand a double banked watch as the Engineering Watchkeeper to the satisfaction of the MSEO. This may require multiple watches before the trainee satisfies the requirement.
002.01.004	Operate propulsion system in semi-automatic control mode.	The Trainee is to stand a double banked watch with the training PO having the OOW do non-delaying speed changes so that the Trainee can react to and input the proper speed changes to the satisfaction of the MSEO. This may require multiple watches before the trainee satisfies the requirement.
002.01.005	Transferring fuel.	The Trainee is to transfer fuel from storage tank to settling tank and then from settling tank to service tank.
002.01.006	Operating Controllable Reversible Pitch Propeller (CRPP) locally.	The Trainee is to operate the CRPP setter motor locally and set proper pitch to at least four speed changes with one going from ahead to astern and from astern to ahead.
002.04	The Trainee shall operate the PG&D to the satisfaction of the MSEO by:	
002.04.001	Changing over electrical generators from the MCR at sea.	The Trainee is to flash up and change over electrical load from on DG to the other at sea.
002.05	The Trainee shall operate main gearing to the satisfaction of the MSEO by:	
002.05.001	Engaging and disengaging the cross connect clutches from the MCC.	The Trainee to engage and disengage both cross connect clutches. The training PO to quiz the Trainee on the requirements to be met before the clutches can be engage
002.05.002	Centralize cross connect clutches from the FER LOP and MCC.	The Trainee to centralize both cross connect clutches after bringing both GTs up to idle. One from the FER LOP and the other from the MCC.

002.05.003	Perform a dead shaft pickup with the PDE in manual from the AER LOP and in auto from the MCC.	The Trainee is to flash the PDE to idle and then do a dead shaft pick up from the AER LOP (local) and MCC (auto)
002.06	The Trainee shall operate the Main Propulsion Engines to the satisfaction of the MSEO by:	
002.06.001	Flash up/shut down PDE in Auto Mode.	The Trainee is to brief the Training PO on the requirements to meet before flashing up the PDE in auto and what to look for before shutting down the PDE.
002.06.002	Flash up/shut down PDE in Manual mode from the PDE LCC and AER LOP.	The Trainee is to flash the PDE up in manual from the PDE LCC and then transfer the control of the PDE to the AER LOP. After successful transfer, the Trainee is to shut down the PDE from the AER LOP.
002.06.003	Perform a detergent/water wash on a propulsion engine to include: 1. Configuring HSS and GT water wash system correctly; and 2. Perform/monitor auto/manual washes.	The Trainee is to direct the Roundsperson on proper PPE requirements before sending them down to configure the water wash system for a detergent wash for the GT. The Trainee is to then do a manual water wash from the MCC and discuss with the training PO what the difference is between manual and auto.
002.06.004	Flash up/shut down GTs in auto and manual from the MCC.	The Trainee is to flash up the GT, once in auto and once in manual. The Trainee is to explain to the training PO the milestones that they are looking for when the GT is starting.
002.06.005	Flash up/shut down GTs in manual mode from the FER LOP.	The Trainee is to do a complete manual start from the FER LOP explaining everything they are doing and looking for to the training PO.
004	The Trainee shall correctly perform delaying drills as Watchkeeper as per reference C-03-050-HFX/MB-02. All drills listed in the manual require a signature. If a drill is added/removed from the manual it is to be automatically added/removed from this OJPR.	
	All the drills below will be assessed by the Training PO.	

Main Lube Oil Failure.			

Main Gearing and Shafting: High Bearing Temperature.			
Propulsion Engine Enclosure Fire.			
Propulsion Engine Lube Oil Failure.			
Propulsion Engine High Lube Oil – Bearing temperature.			
Propulsion Engine Fuel Leak.			
Propulsion Engine Over Speed Condition.			
Gas Turbine High PTIT.			
Gas Turbine High Vibration – Compressor Stall above idle.			
Gas Turbine Hot Start.			
Gas Turbine Internal Fire.			
Gas Turbine High Air Intake Differential Pressure.			
PDE High Fresh Water Temperature.			

CRPP Failure.			
Loss of Main Fuel System Pressure.			
Loss of High Pressure Air.			
Loss of Low Pressure Air.			
IPMS Failure: Station in Control Crash.			
IPMS Failure: Loss of Critical RTU from Network.			
IPMS Failure: Critical RTU Crash.			
IPMS Failure: Secondary RTU Crash.			
Diesel Generator Crankcase: High Pressure Alarm.			
Electrical Power Failure: Blackout at Sea.			
Diesel Generator Enclosure Fire.			
Final set of drills containing no less than three continuous drills conducted by and to the satisfaction of the MSEO.			

ANNEX A – MAIN REFERENCES

Refs Code	Publication Number	Title
A1	C-25-512-000/MS-001	Auxiliary Seawater Circulating System
A2	C-23-HFX-000/MB-001	HALIFAX Class Propulsion Operating Manual
A3	C-27-784-000/MS-001	Low Pressure Compressed Air System Manual
A4	C-27-783-000/MS-001	High Pressure Compressed Air System Manual
A5	C-24-541-000/ME-001	Engine Instruction LM2500 Gas Turbine System
A6	C-24-541-000/MF-001	Schedule Maintenance Engine LM2500 Gas Turbine System
A7	C-27-302-000/MS-001	Main Propulsion Gearing Manual (Vol. 1, Book 1)
A8	C-25-510-000/MS-001	Fuel Oil Filling System Manual
A9	C-24-293-000/MS-001	Main Propulsion System Manual
A10	C-23-HFX-000/MS-002	Canadian Navy Damage Control Information Book for CPF
A11	C-24-547-000/MS-002	Hydraulic Start System for CPF 330 to 335
A12	C-24-547-000/MS-001	Hydraulic Start System for CPF 336 to 341
A13	C-25-509-000/MS-001	Fuel Oil Service System
A14	C-25-550-000/MS-001	Fuel Filter Water Separator
A15	C-26-320-000/MS-001	Primary Electrical Power Generation and Distribution
A16	C-41-541-000/MM-001	LM2500 Gas Turbine System
A17	C-24-545-A00-NY-001	Technical Manual, Main Shaft Flexible Coupling
A18	C-24-545-B00/MS-001	Technical Manual, Main Thrust Block Bearing
A19	C-24-545-000/MS-001	Stern Shaft Seal and Bulkhead Seal
A20	C-24-545-D00/MS-001	Technical Manual, Plummer Bearing
A21	TBD-Drafts	IPMS for HALIFAX Class Equipment (Description)

A22	TBD-Drafts	IPMS for HALIFAX Class Equipment (Data Summaries)
A23	TBD-Drafts	IPMS Operating Instructions Manual
A24	C-29-345-000/MS-001	System Manual, Main Refrigeration
A25	C-27-785-000/MS-001	Fresh Water System
A26	C-27-782-000/NQ-001	Oil Pollution Abatement and Bilge Stripping System
A27	C-27-790-000/MS-001	Sanitary System Manual
A28	C-27-867-000/MS-001	Service Steam System
A29	C-29-355-000/MS-001	Heating, Ventilation and Air Conditioning Manual, Books 1 and 2
A30	C-30-005-033/AA-000	Naval Engineering Manual (NEM), Vol. 1
A31	C-03-050-HFX/MB-002	Manual of Emergency Procedures Marine Systems Engineering (at sea)
A32	C-25-539-000/MS-001	Motor Driven Lube Oil Pump Technical Manual
A33	C-03-010-578/MS-F01	CPF 690222 Moisture Separator System (LM2500)
A34	C-25-547-000/MS-001	OSDA 7-02-366 Lube Oil Separator
A35	C-24-559-000/MS-001	Controllable Pitch Propeller and Propeller Shafting
A36	C-24-548-000/MS-001	20 PA6 V 280 Pielstick Technical Manual, Vol. 1
A37	C-24-559-000/TB-001	Technical Bulletin Marine CRPP Control Oil Cooler Operating Instruction – Applicable to HALIFAX Class Vessel
A38	C-27-778-000/MS-001	Operating and Maintenance Instruction with parts Identification List Steering Gear Control System
A39	C-27-945-000/MS-001	Operation and Maintenance Instruction with Parts Identification List Hydraulic System (Wagner) Steering Gear Control System
A40	C-26-321-000/MS-001	Generator Set Diesel 850KW Vol 1
A41	C-26-321-000/MS-002	Generator Set Diesel 850KW Vol 2
A42	C-27-791-000/MS-001	Sea Water Services
A43	C-24-555-000/MS-001	Technical Manual for Gas Turbine Water Wash Tank
A44	C-26-411-000/TB-001	MWM 602 Technical Bulletin Decoking Instructions

A45	C-23-HFX-000/MS-003	Manual of Trim and Stability Applicable to Halifax Class
A46	N/A	Ship's Standing Orders AL10 (or latest version at time of course)
A47	N/A	CFCD 132 –DC Manual
A48	N/A	Safety and Environmental Management System (SEMS) Halifax Class
B1	B-GN-007-RCN/FP-001	CFCD 132-RCN DC Manual
C1	N00023-80-C2080	Instruction for MLO Relief Valve

ANNEX B – TASK LIST (TRAIN)

Task No	Task Statement	PO
CT0350	Supervise engineering pre-sailing checks	001
HT0040	Operate IPMS man machinery interface	001
HT0070	Operate IPMS equipment health monitoring	001
HT0050	Operate IPMS software utilities	001
HT0060	Operate IPMS applications	001
ALT1520	Test components with built-in test equipment (BITE)	001
AKT1090	Investigate system operation logs	001
AMT0160	Interpret drawings/schematics	001
IT0380	Manage damage control conditions	001
IT0510	Monitor damage control system	001
ET0230	Lead collective operation of auxiliary systems	001
XT0090	Operate steering system remotely	001
DT0120	Operate fuel oil filling/transfer system remotely	001
ATT0920	Operate fluid piping system	001
AIT0870	Test auxiliary seawater circulation automatic sequence function	001
AIT0890	Test auxiliary seawater circulation system emergency supply change over function remotely	001
WT0090	Monitor ship’s stability	001
WT0040	Assess ship’s stability	001
CT0270	Monitor liquid level management system	001
WT0010	Adjust ship’s stability	001
WT0110	Record ship’s stability variables	001
DT0060	Lead collective operation of ancillary systems	001
DT0070	Operate ancillary systems	001
DT0200	Operate main seawater circulation system remotely	001
DT0180	Operate main lube oil system remotely	001
DT0090	Operate the emergency functions of the main lube oil pumps remotely	001
AIT0620	Operate infrared suppression system remotely	001
AVT0770	Operate gas turbine water wash system remotely	001
QT0400	Operate gas turbine anti-icing system remotely	001

MT0020	Operate hydraulic start pack remotely	001
ABT0120	Operate controllable reversible pitch propeller (CRPP)	001
CT0280	Operate generators in all modes	001
CT0060	Configure ship's electrical load	001
ABT0050	Lead operation of gearing and shafting systems	001
ABT0170	Operate main gearing and shafting systems	001
BT0200	Testing operation of gearbox	001
ABT0150	Operate gearbox clutch remotely	001
ABT0080	Monitor shaft line operation	001
ABT0070	Monitor gearbox operation remotely	001
RT0010	Operate main propulsion system in all modes	001
ABT0210	Operate propulsion control remotely	001
QT0420	Operate main propulsion gas turbine remotely	001
QT0390	Operate main diesel engine remotely	001
ABT0200	Operate propulsion control locally	001
QT0410	Operate main propulsion gas turbine locally	001
QT0350	Operate main diesel engine locally	001
GT0010	Advise command on machinery/system status	001
CT0130	Investigate collective operation of auxiliary systems	002
AVT1330	Troubleshoot auxiliary diesel engine internal components	002
AVT1290	Troubleshoot auxiliary diesel engine control system	002
AVT1360	Troubleshoot auxiliary diesel engine sea water cooling system	002
AVT1300	Troubleshoot auxiliary diesel engine exhaust system	002
AVT1350	Troubleshoot auxiliary diesel engine starting air system	002
AVT1320	Troubleshoot auxiliary diesel engine fresh water system	002
AVT1310	Troubleshoot auxiliary diesel engine fuel system	002
AVT1340	Troubleshoot auxiliary diesel engine lube oil system	002
AVT1280	Troubleshoot auxiliary diesel engine combustion air system	002
AHT0830	Troubleshoot auxiliary seawater circulation system	002
AHT0890	Troubleshoot seawater service system (firemain)	002
AHT0840	Troubleshoot bilge system	002
CT0120	Investigate collective operation of ancillary systems	002

AHT0820	Troubleshoot ancillary systems	002
AVT1430	Troubleshoot gas turbine water wash system	002
AVT1450	Verify operating parameters of gas turbine water wash system	002
AHT0880	Troubleshoot main seawater circulation system	002
AHT0870	Troubleshoot main lube oil system (MLO)	002
AHT0850	Troubleshoot main lube oil system filling and transfer system	002
CT0150	Investigate collective operation of power generation and distribution system	002
CT0160	Investigate operation of gearing and shafting system	002
BT0050	Evaluate gearbox function	002
BT0060	Evaluate shaftline function	002
CT0140	Investigate collective operation of main propulsion system	002
CT0430	Troubleshoot main propulsion system	002
AHT0860	Troubleshoot fuel oil system	002
ALT2180	Troubleshoot propulsion diesel engine fire detection control system	002
ALT2170	Troubleshoot propulsion diesel engine enclosure carbon monoxide sensor	002
ALT2200	Troubleshoot propulsion diesel engine low pressure control air	002
JT0040	Advise command on HVAC/CBRN air distribution system capabilities and limitations	003
JT0020	Advise command on chilled water and hydronic system capabilities and limitations	003
JT0030	Advise command on hotel system capabilities and limitations	003
CT0030	Advise command on power generation and distribution system	003
PT0010	Advise command on ship’s power configuration	003
CT0050	Advise command on ship’s electrical load	003
CT0330	Report on power generation and distribution system	003
GT0010	Advise command on machinery/system status	003
CT0040	Advise command on propulsion capabilities and limitations	003
CT0020	Advise command on marine systems operational capabilities and limitations	003
CT0010	Advise command on marine systems engineering parameters	003
CT0340	Report on propulsion deficiencies	003
CT0310	Report on machinery limitations	003
CT0320	Report on marine systems engineering parameters	003
IT0470	Manage propulsion plant operation in action and emergency stations	003
WT0020	Advise command on ship’s stability	003

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates: 6 July – 26 July 2017

Location: WP1020 (Haida –Venture)

Board Members:

Surname	Initials	Rank	MOSID	Unit
Read (Chair)	D.M.	SLt	00345	NFS(P) – DC Div
Robinson (TDO)	T.	Lt(N)	00211	NTDC(P)
Sprang (TDO)	L.A.	Lt(N)	00211	NTDC(P)
TzoTzos (TDO)	L.	Lt	00211	NTDC(P)
Moazzezi Moghaddam (OPI)	A.	Lt(N)	00345	NTDC(P)
Gagne (NPTG Rep)	Y.S.	CPO2	00379	NPTG HQ
Kiervin (Secretary)	S.G.	A/SLt	00344	NFS(P) – CSD
Hockey (SME – MESO)	M.L.	CPO2	00379	F4 ENG
Moody (SME – ET)	M.E.	CPO2	00379	NTDC(P)
Lambert (SME – ER)	S.J.	CPO2	00379	NTDC(P)
Penney (SME – ET)	K.G.	PO1	00379	NTDC(P)
Switzer (SME – HT)	C.M.	PO2	00379	NTDC(P)
Morley (SME – ER)	J	PO2	00379	AP NTDC(P)
Tong (SME – ER)	J.	LS	00379	NTDC(P)

Board Objective:

The purpose of this writing board is to develop the QSP for the USQ-LS Watchkeeper in the overall context of the Marine Technician trade development MIP. This is the third QSPWB to be convened in the process.

Specification Task – No Train Decisions:

Decisions are captured as comments in the JTAR where an NT occurred. NT tasks were usually decided based on redundancy or belonging at another level such as NETP.

The NT tasks decided during the board are as follows:

Task Number	Task Description	Reason
QT0350	Operate Main Diesel Engine (MDE) Locally	OS Level Training
QT0360	Operate Main Diesel Engine (MDE) Manually	OS Level Training
DT0140	Operate Fuel Oil System Locally	OS Level Training
DT0150	Operate Fuel Oil System Remotely	OS Level Training
ET0140	Operate Bilge Eductor Locally	OS Level Training
AST0020	Adjust POL in Marine Systems and Equipment	Roundsperson Training
ATT0010	Adjust Fluid Piping System (Steam, Water, Aux, Circulation, Chilled Water, Fuel, Air)	OS Level Training
QT0280	Operate Auxiliary Gas Turbine Remotely	Lack of Relevant System
AST0020	Adjust POL in Marine Systems and Equipment	OS Level Training
QT0380	Operate Main Diesel Engine (MDE) pre-lube oil system remotely	OS Level Training
AOT0960	Operate Black Water System Remotely	Lack of Relevant System
QT0320	Operate Diesel Engine Pre-Heat System Remotely	OS Level Training
ET0120	Operate Auxiliary Steam Generator Remotely	OS Level Training
ET0100	Operate Auxiliary Seawater Circulation System Remotely	OS Level Training
CT0290	Operate Ship’s Electrical Load	OS Level Training
AUT1950	Parallel Electrical Loads Remotely	OS Level Training
CT0270	Monitor a Liquid Level Management System Console	Lack of Relevant System
CT0080	Configure Static Frequency Converter	OS Level Training
OT0140	Monitor Ship’s Liquid Cargo System	Lack of Relevant System
ALT0880	Operate Battle Damage Control System	Prior Experience

CT0100	Coordinate Engineering Pre-Sailing Checks	LS Tech
AIT0900	Test System Operation of Fuel Oil Filling Transfer System	Prior Experience
XT0060	Operate Steering Control Remotely	OS Level Training
DT0080	Operate Emergency Functions of Main Lube Oil Pump Locally	LS Tech (Mech)
ABT0140	Operate Gearbox Clutch Locally	LS Tech (Mech).
BT0090	Inspect Main Propulsion Shafting	Prior Experience
DT0170	Operate Main Lube Oil System Locally	LS Tech (Mech).
ABT0040	Inspect Visually Main Shaft Line	Prior Experience
BBT0540	Monitor ship's fluid tanks	OS Level Training
AOT0090	Adjust space ventilation	OS Level Training

Discussions/Decisions:

05 Jul 17 Introduction Brief was conducted by TDO to familiarize everyone with the QSWB expectations and the outline for how the board will run. Many of the board members had also participated in the USQ AB QSPWB. The Board Chair also provided some background on his career, how disputes will be handled, and what the timings will be. After intro brief, the Review Document brief was given. The board developed a definition of the LS Watchkeeper and the expectations of the position WRT each class of ship. The classes considered included ORCA, KINGSTON, and HALIFAX class ships. Of specific interest was the legacy console course and how to align future training with the capability required of current personnel. Questions remained following this discussion WRT tasks required on the ORCA class ship.

Since most members of this board participated in the previous AB board the members were then instructed to discuss lessons learned from previous QSPs and which aspects were successful or not. This was done IOT improve the quality of the product from this board as well as the actual board experience for the members. Points which were discussed included:

- EC/PC When not required (Avoid unnecessary testing)
- TDO inconsistency (Resolved)
- More OJT (Apprenticeship)
- Remember IMO
- File Management, specifically naming convention (Proposed: PO 002 Boilers 27/06/17)

- Let SMEs do their job (WRT higher level command decisions)

- Move forward with training task list as specified by TDO only

The previously collected No Train tasks were then reviewed IOT ensure that none should actually be included in LS Watchkeeper training. It was confirmed that none of the currently designated NT tasks should be reclassified as training tasks.

Following a visit from the NPTG Training Development Chief, CPO2 Gagné, it was decided that Section Base IC on KINGSTON training should be included at the tech level

Attack team leader course was incorporated into AB USQ, duty watch tasks are considered pre-requisites.

Next, the board reviewed tasks currently designated as Train WRT the three relevant classes of ship only. This was done IOT ensure none of the training tasks were redundant. This was the first step toward grouping the tasks according to component and creating scalars. The tasks were initially organised into Ancillary, Gearing, Propulsion, Stability, Operate Diesel, Auxiliaries, Pre-Watch Rounds, Advise Command, Emergencies, IPMS, and No Train.

At this time it was decided that the legacy console course must be included into the Watchkeeper training. It was clarified that the console course involved learning IPMS and was a cert 2 requirement.

The broad structure of the Watchkeeper course was discussed between a general course followed with more specific class packages or three separate QSPs, and any option in between. Currently, conversion training is used to transfer members between different classes of ships at a certain point in their training.

Further discussion related to whether the Watchkeeper requires ‘tech’ training, meaning that they have a deeper understanding of how their systems work, versus only responding in terms of defined drills. This was relevant to defining which tasks would be required training. This discussion lead to the statement that OJT would be fundamental to achieving competency in these tasks. However, this was countered with options for simulation instead. The general sentiment was that OJT includes necessary real life experience. This was refined to consider that only testing would be necessarily conducted on OJT.

The phrase Troubleshoot was changed to recognise or identify, ideally it is stated that these tasks would continue into further training at the LS Tech level.

It is reiterated that all Watchkeeper tasks must remain Level 3 tasks (Independently Performed) due to the nature of the position.

06 Jul 17 SMEs verified the tasks to be trained at LS level in the Watchkeeper course using JTAR. Then, the team broke into different groups to re-scalar the POs and EOs. Some of the tasks are to be repeated for different tech courses to align with career progression of Mar Techs.

The board discussed the options to train Mar Techs on the briefing aspect of the watchkeeping. One option is to introduce a common phase before the specialty training. The other is to separate them into class packages to ensure the knowledge tailored to specific class vessels.

The board discussed whether or not monitoring should be covered under 'briefing the commander'. Then the board discussed on the extent of the verb *to monitor* and *to troubleshoot*.

07 Jul 17 The board discussed options to train students on the tasks for systems such as IPMS and IPCS. First option is to focus on the system. OS and AB level QSPs have been developed this way. Second option is to focus on the processes. Expectations of a Watchkeeper are different from OS and AB. There are portions of the systems already covered in duty watch package. They involve responding to emergencies.

The board discussed the viability of the process-based approach. Systems that can't be taught by simulators will be taught during OJT at sea.

The board tried producing scalars using the process-based approach. The board created the example scalar based on reacting to emergencies.

"Advise" tasks have been distributed under *Operate* and *Respond* based on the board members' discretion. The *Respond* verbs were related to emergency.

Tasks CT0280 and CT0060, "Operate generators in all modes" and "Configure ship's electrical load" respectively, have been moved back to trained tasks during the scalaring process. This was done because they were redefined as necessary Watchkeeper capabilities

With regards to exams, the board decided that the PCs should not be carbon copies of the ECs. And they should be randomized scenarios to add complexity in comparison to PCs.

If failing of an EC would pose safety/security concern, were the member to be employed on ship, the EC should be deemed a critical EC (CEC).

It is implied that the Watchkeeper will be carrying out his/her duty at sea.

The task “CT0150”, “investigate collective operation of power generation and distribution system” has been placed back into the QSP IOT incorporate drills in Watchkeeper training.

The Watchkeeper on HFX class vessel would have less authority in the MCR. Therefore, Sr Watchkeepers are the ones briefing the command in HFX class vessels. There are to be two personnel in the MCR, so it is advisable to push the “advise” task towards the Sr Watchkeepers (HFX only).

The board has decided to incorporate SHINCOM training into the QSP to produce more proficient Watchkeepers at the LS level because there are pipes unique to different situations.

Prior to adjourning of the meeting, the board commenced drafting performance objectives for Chapter 2 for Halifax class vessels.

Abbreviations used in Chapter 2 have been added to the list in Annex F.

10 Jul 17 The board reviewed and edited Chapter 2 of POs 001 through 004 for HFX class ships drafted last on 07 JUL 17.

Next, the board moved to construct the same four POs for the KINGSTON class ships. A notable difference was the change from the IPMS console to the MCR console.

Following this the board began constructing the four POs for the ORCA class ships. Discussion focussed on the difference in manning on an ORCA. Of particular interest was the added duty of Fueling IC required of the ORCA class LS Watchkeeper. This lead to discussion regarding the greater scope of capabilities required of an ORCA class engineering Watchkeeper, such that the position might be so involved as to require a LS Technician instead. Lt(N) Lesieur advised the board that this greater scope will likely need to be included at the Watchkeeper level. It was clarified that the Watchkeepers who move from one class of ship to another would take the corresponding on board watchkeeping course for that class. Further concerns existed regarding duplication of training. This discussion was put on hold.

Instead, the board continued to draft Chapter 1 for the HFX class ships. Discussion ensued regarding the necessary prerequisites for the Watchkeeper course. Of specific concern was whether a class specific Roundsperson package would be required versus a general Roundsperson package. Requiring a class specific Roundsperson would mean for example that a KIN class qualified Watchkeeper would need to requalify from the Roundsperson level were they transferred to a CPF. It was decided that an OJT could be used to convert qualified Watchkeepers from class to class.

The board then drafted Chapter 1 for the KIN class ships. During this draft a discussion regarding modularization of the courses arose. This is considered very important for reservist training. However, the course is likely too short to be appropriate for modularization. Finally, discussion arose WRT whether the LS rank should be a pre-requisite as well.

The NPTG Training Development Chief, CPO2 Gagné was in attendance to offer guidance regarding earlier discussions which had been placed on hold (Conversion, and ORCA class). IOT resolve the ORCA class discussion it was decided that the OCE QSP package would be acquired for study.

Additionally, it was decided that some tasks need to be added in to properly define the LS Watchkeeper position. This will be a very involved process and proper confirmation that appropriate tasks do not already exist will be conducted.

The OA did not provide appropriate tasks for the following:

- Operate Communications System
- Monitor Auxiliary System
- Monitor Main Propulsion
- Respond to Ancillary System Alarms
- Respond to Gearing and Shafting Alarms

Some concern was raised regarding the steep learning curve of the required tasks, it was clarified that the PO's don't need to be taught in sequence. This means they can be organized to allow for better instruction. This conversation worked back into discussion about modularization and testing. Decisions WRT modularization will continue as the length of lessons is solidified.

The board then moved into the 'design phase' the steps of which include: target population analysis (TPA), instructional analysis, enabling objectives, assessment plan, costing, and instructional strategy.

The TPA outlined the following:

- MAR TECH
- Grade 10 Math
- Basic Mil Exp
- All ages (Within Limits of Employment)
- At least AB
- Legacy Trade

The board then reviewed the previously designated NT tasks in order to determine if any could viably be returned to T as applicable to the capabilities which currently require tasks to be added (As listed earlier).

Thus, it was determined that ABT0100 Operate Central Alarm Monitoring System should be trained under ‘Operate Machinery Console’.

Then, the board worked on scalaring tasks for the HFX class so they could move into the composition of Chapter 4. The important distinctions in this step were whether tasks were independent teaching points, dependent, or ‘to include’ subjects.

The board proceeded to begin writing the EOs for each of the four POs for each of the classes of ship. Continuing to postpone the ORCA class composition.

The tasks Monitor a Liquid Level Management System (LLMS) Console (CT0270), Monitor Ship’s Fluid Tanks (BBT0540), and Monitor Liquid Cargo System (OT0140) were added to monitor auxiliary systems (PO 002) as T tasks from the NT list.

11 Jul 17

Lt(N) T.Robinson, TDO

Lt(N) L.A Sprang, TDO

SLt D. Read, Board Chair

Lt(N) A. Moazzezi Moghaddam, OPI

A/SLt S.G Kiervin, Secretary

CPO2 M.L Hockey, MESO SME

CPO2 S.J. Lambert, ER SME

PO2 C.M. Switzer, HT SME

LS J.T. Tong, ER SME

MS C. Osborne, ET SME

PO1 K. Penney, ET SME

The NPTG Training Development Chief, CPO2 Gagné communicated that the OCE course would replace the LS Watchkeeper course for ORCA class ships. The board will not be required to create this document.

The board began drafting each EO individually to be compiled in the master HFX USQ document afterward. Once this was completed the EOs were reviewed collectively. When reviewing EO 001.07 Operate Communication Systems it was evident that necessary tasks might need to be added so discussion as to possible solutions arose. It was decided that other tasks could be redefined or moved around. The board moved the task Advising command on machinery/system status (GT0010) into the EO IOT satisfy this need.

Moving into reviewing the EOs for PO 002 (Monitor) started discussion about moving this PO into PO 003 (Investigate) or PO 001 (Operate). This would solve the issues regarding task creation under PO 002. Some tasks moved into PO 001 (Operate) and some were moved into PO 003 (Investigate) where applicable. Thus PO 003 (Investigate) became PO 002 (Investigate) and PO 004 (Respond) became PO 003 (Respond).

The board then reviewed the EOs in PO 001, PO 002, and PO 003.

12 Jul 17

Lt(N) T.Robinson, TDO

SLt D. Read, Board Chair

Lt(N) A. Moazzezi Moghaddam, OPI

A/SLt S.G Kiervin, Secretary

CPO2 M.L Hockey, MESO SME

CPO2 S.J. Lambert, ER SME

PO2 C.M. Switzer, HT SME

LS J.T. Tong, ER SME

MS C. Osborne, ET SME

PO1 K. Penney, ET SME

PO2 J. Morley, ER SME

The board began with a breakdown of what a normal training day schedule should look like on the LS Watchkeeper course. Based on a given 400-minute day it was understood that there would be 8 periods in a day. This was useful for the next task the board would conduct of drafting the teaching points for each EO of the HFX class Watchkeeper package. The EOs were divided among the board members to complete individually.

It was decided that the tasks Monitor ship's fluid tanks (BBT0540) and Monitor Liquid cargo system (OT0140) should be NT tasks because they were repeated in other tasks.

13 Jul 17

Lt(N) T.Robinson, TDO

SLt D. Read, Board Chair

Lt(N) A. Moazzezi Moghaddam, OPI

A/SLt S.G Kiervin, Secretary

CPO2 M.L Hockey, MESO SME

CPO2 S.J. Lambert, ER SME

PO2 C.M. Switzer, HT SME

LS J.T. Tong, ER SME

MS C. Osborne, ET SME

PO1 K. Penney, ET SME

PO2 J. Morley, ER SME

It was revealed to the board that rank cannot be considered a prerequisite for training. This meant that the ‘LS Watchkeeper’ could in fact not be a LS.

Next, the board continued to work on the teaching points for the HFX class Watchkeeper package. Simultaneously the EOs for the KIN class package were developed in accordance with the KIN class specific task list. Once some EOs had been completed, the board reviewed them.

During this review, it was clarified that the length of time dedicated to teaching points should be considered in terms of minutes and not portions of a school period. The length of time for each lesson was of particular interest as well as the portion of time which would be dedicated to either practical learning or theoretical study.

Next, the board reviewed the list of references which would be applied to each of the EO’s teaching points. The base list of references was drawn from the Machinery Control Console (MCC) and revised by the board as necessary.

EOs 1 through 7 for PO 001 had been reviewed at close of business.

14 Jul 17 The board was otherwise occupied.

17 Jul 17

Lt(N) T.Robinson, TDO

SLt D. Read, Board Chair

A/SLt S.G Kiervin, Secretary

CPO2 M.L Hockey, MESO SME

CPO2 S.J. Lambert, ER SME

PO2 C.M. Switzer, HT SME

LS J.T. Tong, ER SME

MS C. Osborne, ET SME

PO1 K. Penney, ET SME

PO2 J. Morley, ER SME

The board discussed a point from 13 Jul regarding the lack of a rank based pre-requisite. It was decided that a replacement pre-requisite should be added requiring 400 hours on watch. This was done so that the incoming students have some experience. The board however still felt that the rank pre-requisite should remain. This issue was placed on hold for further review.

Then, the board continued to draft the teaching points for EOs in both the HFX and KIN class ships. Paragraph 10 of each EO in the HFX class package were reviewed.

18 Jul 17

Lt(N) T.Robinson, TDO

SLt D. Read, Board Chair

A/SLt S.G Kiervin, Secretary

CPO2 M.L Hockey, MESO SME

CPO2 S.J. Lambert, ER SME

PO2 C.M. Switzer, HT SME

LS J.T. Tong, ER SME

MS C. Osborne, ET SME

PO1 K. Penney, ET SME

PO2 J. Morley, ER SME

The board began with discussion regarding the level of experience required to conduct the Watchkeeper position. This was done to clarify if the ‘400 hours’ type of experience could be satisfactory, or if the rank pre-requisite should be enforced, or if possibly the LS tech course should be mandated prior to LS Watchkeeper.

The board began working on paragraphs 4 through 9 of each EO in the HFX class package. Simultaneously discussion focussed on tests which should be considered critical evaluations or not. The purpose being to alleviate any fears that the LS Watchkeeper were not prepared for their position.

The current instructor of the existing console course PO2 Sleeman attended the board IOT advise on the trainer’s capabilities and relevant systems. WRT certain systems that couldn’t be simulated on the trainer it was mentioned that the QSP could suggest alternative options for drills. His advice was especially useful for developing timings for instruction as well as testing of EOs. Current instructors of the legacy Mar Tech training relevant to the board should be brought in as consultants during this portion of the board. The board was able to complete and review paragraphs 4 through 9 for all the EOs of the HFX class package.

19 Jul 2017

Lt(N) T.Robinson, TDO

SLt D. Read, Board Chair

A/SLt S.G Kiervin, Secretary

CPO2 M.L Hockey, MESO SME

CPO2 S.J. Lambert, ER SME

PO2 C.M. Switzer, HT SME

LS J.T. Tong, ER SME

MS C. Osborne, ET SME

PO1 K. Penney, ET SME

PO2 J. Morley, ER SME

Discussion regarding critical ECs cannot be tested in the PC so as a result critical ECs are to be changed to EC and a main PC created for PO 002 and PO 003. The board reviewed completed Chapter 4 items for the Kinston class QSP. If applicable one reference list will be used for both packages. The board discussed the MCR trainer. Two of the tasks have been taken out from EO 001.03 KIN during the production of the draft. DT0180, and DT0090. These tasks are still being trained, but removed from the KIN QSP due to their irrelevance.

The board then discussed with Lt(N) Lesieur the total length of the course and the number of training days required. This then lead to discussion regarding the necessary addition of PT periods and whether that applies to a course of this length (~23 training days).

Next, the board discussed how to draft ECs based on a template. Of interest in this template was that the ECs will be evaluated only on a pass/fail basis. The board was then directed to compile checklists for the ECs evaluation as well as detailed instructions for the evaluators. It was also mentioned that 'Safety fails' should be included. This meant that certain safety infractions could cause a student to fail the EC immediately.

The board then reviewed the format of chapter 3 of the USQ. Within this review discussion arose regarding paragraph 5 and the responsibility for testing. The board had concerns about conflict of interest where course staff would be testing the same lessons they had taught. Further clarification would be sought. Another concern was whether career managers would be properly advised WRT student's performance and chosen career path. It was decided that this would be managed by the existing infrastructure.

At this point the following changes from the initial convening order should be recognised:

The board met from 5 Jul to 26 Jul instead of 10 Jul to 4 Aug.

This excluded 14 Jul where the board members were required on parade.

CPO2 M.E. Moody was on leave until Jul 18 and did not meet with the board.

PO1 K.G. Penney was absent for two afternoons due to parade practice.

PO1 D.I. Lester was re-tasked as of 10 Jul.

MS R.W. Medal was re-tasked as of 7 Jul.
LS C.G. Welch was re-tasked as of 7 Jul.

20 Jul 2017

C-14

Lt(N) T.Robinson, TDO

SLt D. Read, Board Chair

A/SLt S.G Kiervin, Secretary

CPO2 M.L Hockey, MESO SME

CPO2 S.J. Lambert, ER SME

PO2 C.M. Switzer, HT SME

LS J.T. Tong, ER SME

MS C. Osborne, ET SME

PO1 K. Penney, ET SME

PO2 J. Morley, ER SME

The board agreed that there will need to be two separate reference lists for the HFX and KIN class respectively.

It was also agreed that the ‘conflict of interest’ mentioned WRT paragraph 5 of Chapter 3 was not of great concern and that it could be left as is.

The board then continued to draft ECs and PCs for the HFX class package as well as EOs for the KIN class package.

21 Jul 2017

Lt(N) T.Robinson, TDO

SLt D. Read, Board Chair

A/SLt S.G Kiervin, Secretary

CPO2 M.L Hockey, MESO SME

CPO2 S.J. Lambert, ER SME

LS J.T. Tong, ER SME

MS C. Osborne, ET SME

PO2 J. Morley, ER SME

The board discussed the necessity to outline the specific points which are different between a trainer simulation and a real ship at sea. This was important for justifying the need to add an OJT package to the USQ versus only simulator training.

Next, the board reviewed the remaining EOs in the KIN class package for paragraphs 1 through 3, and 10. The board then created a course history for each class package.

24 Jul 2017

Lt(N) T.Robinson, TDO

SLt D. Read, Board Chair

A/SLt S.G Kiervin, Secretary

CPO2 S.J. Lambert, ER SME

PO2 C.M. Switzer, HT SME

LS J.T. Tong, ER SME

MS C. Osborne, ET SME

PO1 K. Penney, ET SME

The board began with a review of the annexes for the USQ. It was noted that course training efficiency could be increased with additional PCOT and MCR trainer workstations.

This review involved a calculation of the training time required as well as other admin time requirements which was used to populate tables in the annexes as well as chapter 1.

25 Jul 2017

Lt(N) T.Robinson, TDO

SLt D. Read, Board Chair

A/SLt S.G Kiervin, Secretary

CPO2 M.L Hockey, MESO SME

CPO2 S.J. Lambert, ER SME

PO2 C.M. Switzer, HT SME

LS J.T. Tong, ER SME

MS C. Osborne, ET SME

PO1 K. Penney, ET SME

The board began with an electronic review of the HFX class package. The first changes made were to clarify that time would be required surrounding PT periods for changing, showering, and travel to the gym. It was decided that this should be very clearly stated so scheduling would be translated well.

Next, the board composed the course staffing table in Chapter 1. It was found that clarification would be required WRT the qualification required for the instructors. The board reviewed paragraphs 4 through 9 of the KIN class EOs and compiled them into the master KIN class package.

26 Jul 2017

Lt(N) T.Robinson, TDO

SLt D. Read, Board Chair

A/SLt S.G Kiervin, Secretary

CPO2 M.L Hockey, MESO SME

CPO2 S.J. Lambert, ER SME

LS J.T. Tong, ER SME

MS C. Osborne, ET SME

PO1 K. Penney, ET SME

The board reviewed the ECs and PCs for the KIN class package. Annexes G, and K, were edited and annex L was removed.

It was also determined that instructors for the course would ideally possess the following qualifications: IPMS OBTS & BDCS Scenario Building 119167-0006, but that IPMS Tech would be the minimum qualification. This was specified due to the lack of current instructors with the ideal qualification.

The board discussed the difference in how to apply the terms of the EOTIs in KIN versus the HFX class ships. The KIN class ships do not have their own engineering officer. The drill assessments must reference the established checklists.

Annex K in the HFX class package was updated and reviewed as well as the OJPR checklist. This involved the addition of an oral board to the list. It was noted by Chief Gagne that the board should not involve a written test nor drawings as per previous instructions. However the board felt that an oral board would satisfy these constraints while achieving a high standard of training.

Annexes E & F were supplied by Chief Hockey for the KIN class package and the board collectively reviewed the entire KIN class document. During this review, it was noted that for both class packages a CF1110[1] should be drafted for all members upon completion of the USQ. This was done to ensure that the member's progress is well tracked. Finally, it was clarified that the instructional staff must develop the test material as well as administering the tests. This was done IOT ensure that the board members' duties were clear and complete.

[1] CF1110 shall not be used for Mar Tech qualifications, NPI 27

The board adjourned with some work left in additions to the Oral Board checklist and the KIN class document. These changes would be completed in the days following the board such that each class package would be ready for review by 02 AUG.

Lessons Learned:

See above comments

Guidance to Instructional Staff:

Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1			
2			

Other Considerations (if applicable):

APR 30

SLt R. Barnes

The following changes were made to this QSP all headings were updated to be consistent throughout the document. All EOs and POs were formatted to start on their own page. Removed all EC that were empty and recommended to be removed. Deleted EO 002.07 as it was considered redundant and was a recommended comment and was slated to be deleted. Changed the following numbers:
EO 001.07 to EO 002.07
EO 004.04 to EC 004.04

Lt(N) Wilson: A final proofread was conducted. The notation under ‘training limitations’ noting that schools may not have the requisite training equipment was changed to ‘nil’ and a note placed in the ‘resources’ section of Chapter 1 directing schools to submit TDRs or alter schedules for instances where the required resources were not available. Additionally, several numbers were cleaned up (e.g. changing EO numbers to match the chapters where it was clearly a typo).

May 1 - 3

LT(N) R. Wilson

SLt R. Barnes

CPO2 S.I. Blackbourn

PO2 C. Gillies

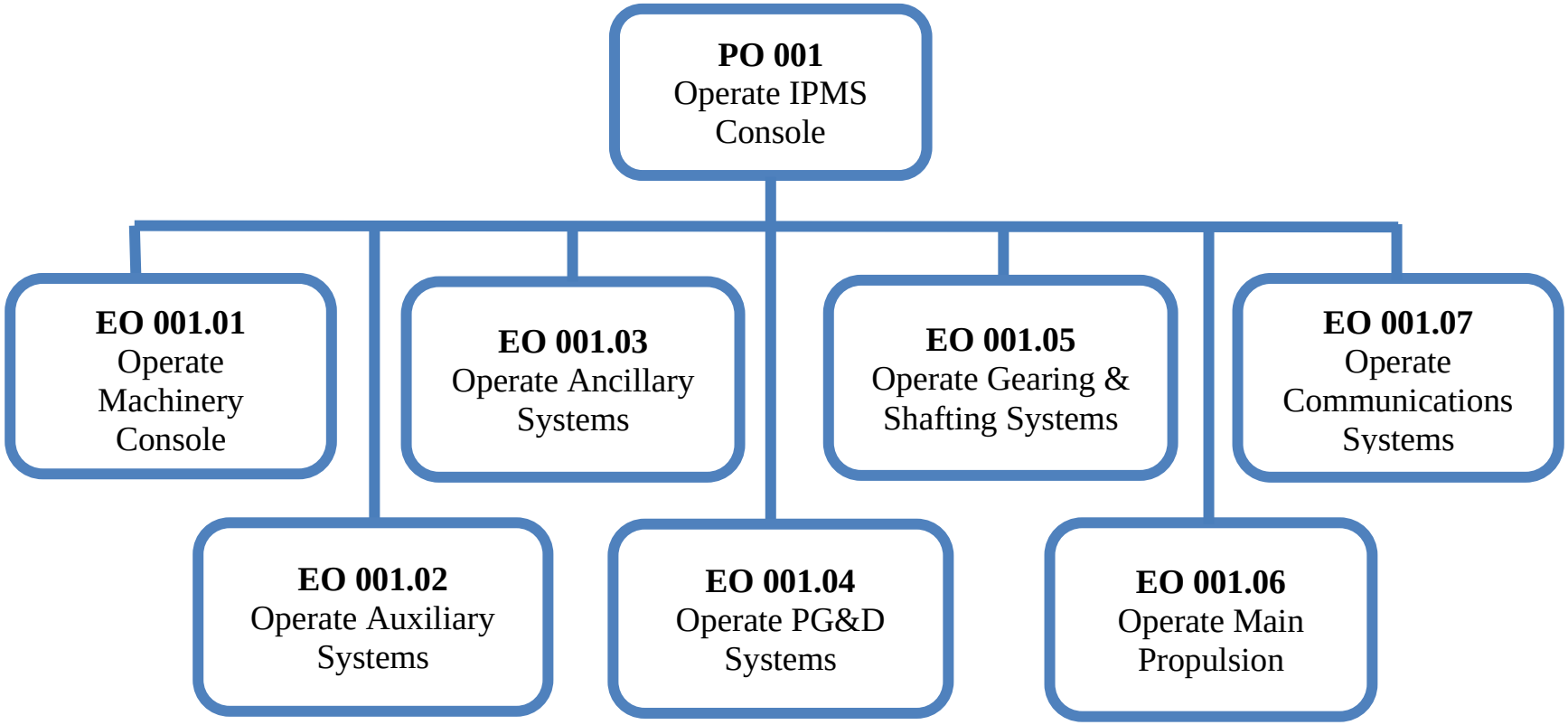
Conducted a cleaning pass to correct numbering schemes, training timings, language, and grammar. PO 001 has been added to correct system knowledge deficiency and formatting was adjusted accordingly. Proper title of “Engineering Watchkeeper” added throughout to replace LS Watchkeeper or similar. Further updates and clarification added, including specifying number of drills in EC 004.04, specifying sea time requirement and removing the drill question from the qualification board. Greater fidelity in the ‘direction to board’ was added.

23 Sep 2022

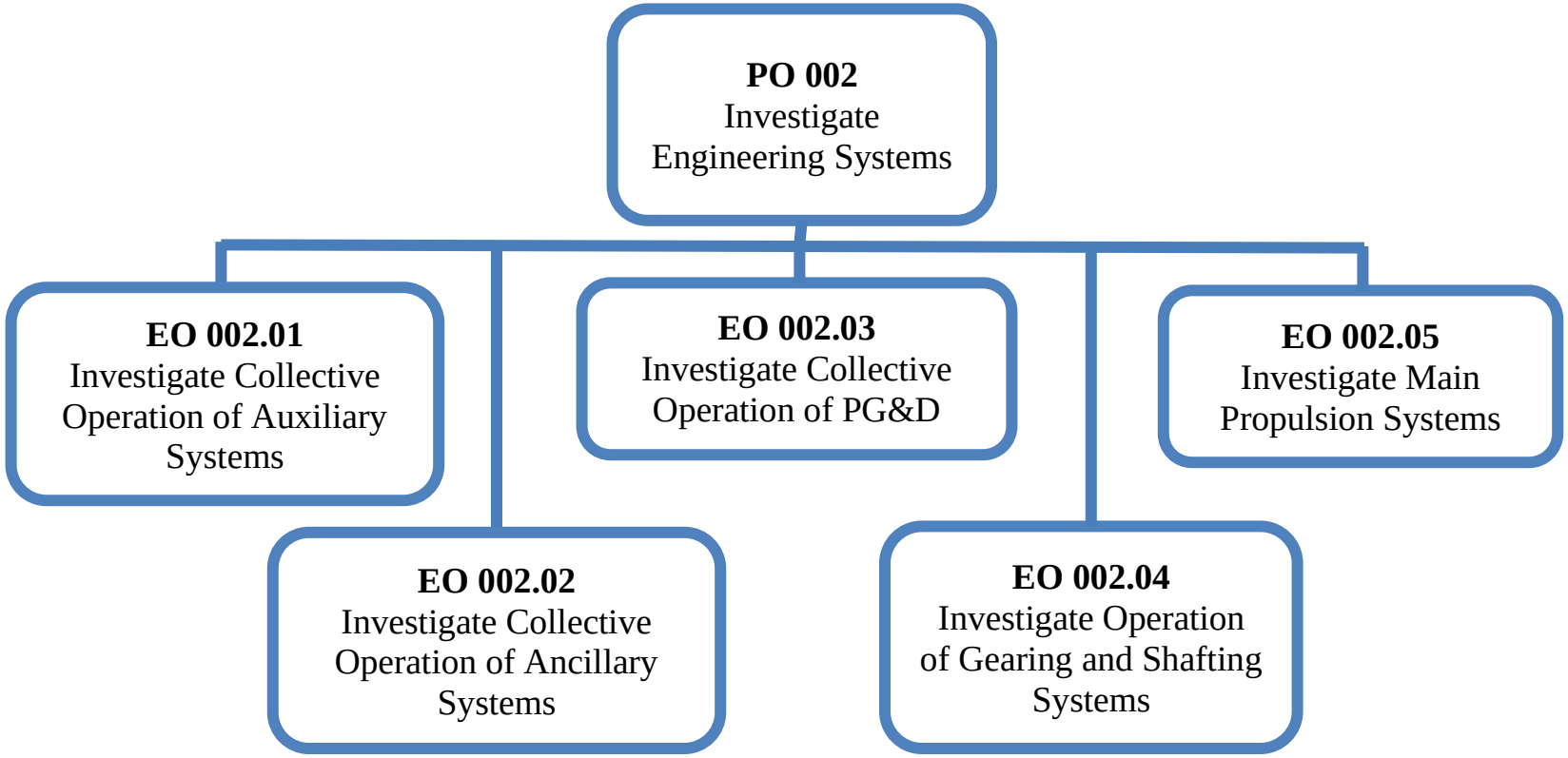
CPO2 M.R. Charland

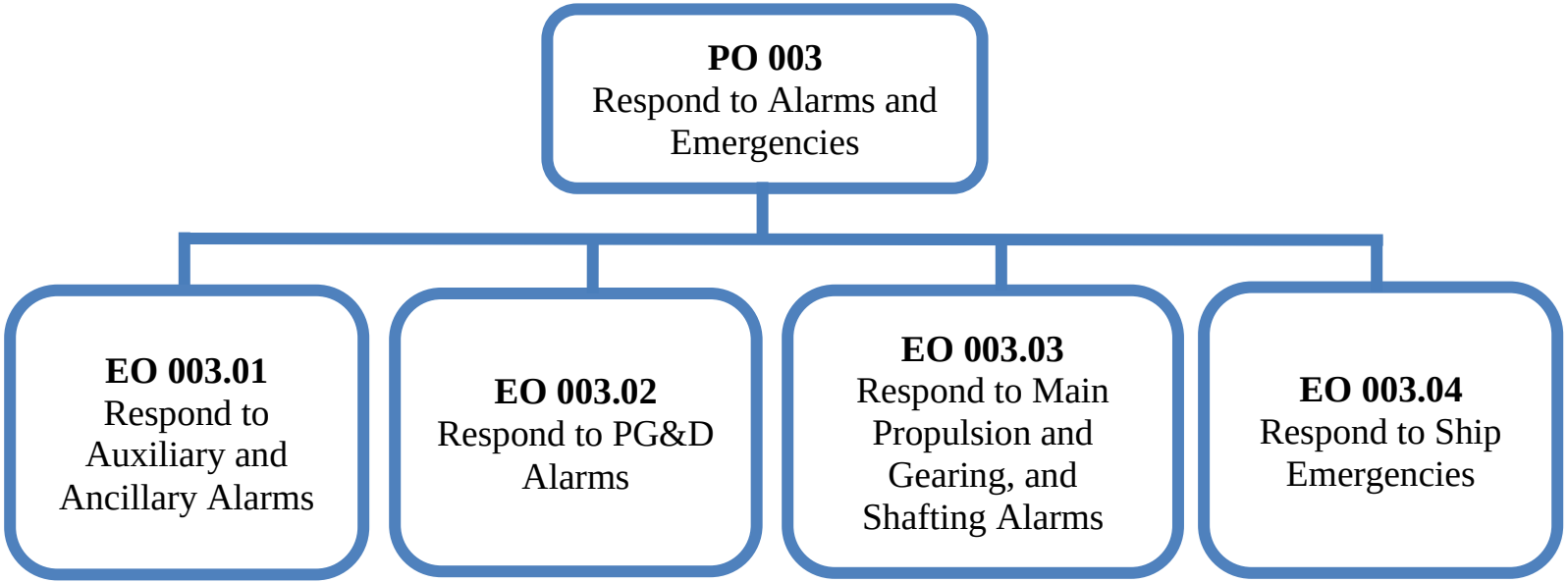
Conducted a review to insure all important information required to be in QSP is present. We made sure all PO and EO have at least 8 recommended paras by the TDO (Performance, Conditions, Standard, References, Task Numbers, Training Limitations, Remarks and Test Details) and we verified if the QSP contained the minimum required annexes (References, Suggested sequence and timings, Assessment plan and Record of proceedings/Decisions). Also due to the removal of the Operation Honour question, and the TM approval, we added a Safety/Environmental replacement question.

ANNEX D - SCALARS



Endorsed: June 2018
Version No: 3.5
MITE Corse ID: 121632-121633


PO 002
Investigate Engineering Systems
EO 002.01
Investigate Collective Operation of Auxiliary Systems
EO 002.02
Investigate Collective Operation of Ancillary Systems
EO 002.03
Investigate Collective Operation of PG&D
EO 002.04
Investigate Operation of Gearing and Shafting Systems
EO 002.05
Investigate Main Propulsion Systems


PO 003
Respond to Alarms and Emergencies
EO 003.01
Respond to Auxiliary and Ancillary Alarms
EO 003.02
Respond to PG&D Alarms
EO 003.03
Respond to Main Propulsion and Gearing, and Shafting Alarms
EO 003.04
Respond to Ship Emergencies

ANNEX E – ON JOB PERFORMANCE RECORD (OJPR)

(Certificate of Qualification)

_____

Name and Initials Rank SN

_____

Date Training Started Expected Date of Completion

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. Recording of the individual's progress shall be entered in Monitor Mass under the Training Tracker. The NFS for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance Objective to the required standard, the MSE Training PO will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and enter progress in the Monitor Mass Training Tracker (MMTT).

3. If it is not possible to conduct OJPR for some POs due to equipment not fitted or not issued (e.g. Work Period), the MSE Training PO is directed to contact Training Manager NPTG HQ for guidance. Extension of training time might be required.

4. See Appendix 1 for OJPR sign off list.

5. Appendix 1: Performance Checks require the trainee to demonstrate the listed competencies as described in Chapter 4. A series of confirmation questions will be asked from each EO to demonstrate the knowledge of the trainee before granting qualification. The trainee will have six (6) months from being posted to an operational ship to complete Appendix 1.

6. On completion of this OJPR, the MSE Training PO shall check box PO 006 in MMTT. This will update automatically Monitor Mass, MPRR and auto generate a message.

7. If you wish to make a change, send your request to: +NPTG MITE@NPTG HQ@Esquimalt

PROGESS REVIEW

8. Progress review shall be carried out monthly by the Divisional Officer and CHOD.

OJPR Reviewed By (Rank, Position, Name)	Signature	Date

TRAINING COMPLETION CERTIFICATE

Date training completed: _____

_____

MAR TECH Chief of Department (CHOD) (Print and sign) Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed training to the standards stated in the Marine Technician Engineering Watchkeeper HFX Class Package. The member is qualified to carry out the duties of a Marine Technician Engineering Watchkeeper and is (MITE 121633) qualified.

_____

Engineering Officer (Print and Sign) Date

ON-JOB PERFORMANCE RECORD SHEET

1. The remarks column in Appendix 1 shall be used for the following information only:

   a. The unit in which the training and assessment were carried out; and

   b. Any unusual circumstances which preclude compliance with an objective.

APPENDIX 1

PO/EO	Performance Statement	Signature / Comments
002.01	The Trainee shall Conduct Watchkeeping operations to the satisfaction of the MSEO by::	
002.01.001	Conduct pre-watch and on-watch rounds.	
002.01.002	Conduct flash-up and shut-down routine.	
002.01.003	Operate propulsion system in automatic control mode.	
002.01.004	Operate propulsion system in semi-automatic control mode.	
002.01.005	Transferring fuel.	
002.01.006	Operating Controllable Reversible Pitch Propeller (CRPP) locally.	
002.04	The Trainee shall operate the PG&D to the satisfaction of the MSEO by:	
002.04.001	Changing over electrical generators from the MCR at sea.	
002.05	The Trainee shall operate main gearing to the satisfaction of the MSEO by:	
002.05.001	Engaging and disengaging the cross connect clutches from the MCC.	
002.05.002	Centralize cross connect clutches from the FER LOP and MCC.	

001.05.003	Perform a dead shaft pickup with the PDE in manual from the AER LOP and in auto from the MCC.	
002.06	The Trainee shall operate the Main Propulsion Engines to the satisfaction of the MSEO by:	
002.06.001	Flash up/shut down PDE in Auto Mode.	
002.06.002	Flash up/shut down PDE in Manual mode from the PDE LCC and AER LOP.	
002.06.003	Perform a detergent/water wash on a propulsion engine to include: 1. Configuring HSS and GT water wash system correctly; and 2. Perform/monitor auto/manual washes.	
002.06.004	Flash up/shut down GTs in auto and manual from the MCC.	
002.06.005	Flash up/shut down GTs in manual mode from the FER LOP.	
004	The Trainee shall correctly perform delaying drills as Watchkeeper as per reference C-03-050-HFX/MB-02. All drills listed in the manual require a signature. If a drill is added/removed from the manual it is to be automatically added/removed from this OJPR.	
	All the drills below will be assessed by the Training PO.	
Main Lube Oil Failure.		
Main Gearing and Shafting: High Bearing Temperature.		

Propulsion Engine Enclosure Fire.	
Propulsion Engine Lube Oil Failure.	
Propulsion Engine High Lube Oil – Bearing temperature.	
Propulsion Engine Fuel Leak.	
Propulsion Engine Over Speed Condition.	
Gas Turbine High PTIT.	
Gas Turbine High Vibration – Compressor Stall above idle.	
Gas Turbine Hot Start.	
Gas Turbine Internal Fire.	
Gas Turbine High Air Intake Differential Pressure.	
PDE High Fresh Water Temperature.	
CRPP Failure.	
Loss of Main Fuel System Pressure.	

Loss of High Pressure Air.	
Loss of Low Pressure Air.	
IPMS Failure: Station in Control Crash.	
IPMS Failure: Loss of Critical RTU from Network.	
IPMS Failure: Critical RTU Crash.	
IPMS Failure: Secondary RTU Crash.	
Diesel Generator Crankcase: High Pressure Alarm.	
Electrical Power Failure: Blackout at Sea.	
Diesel Generator Enclosure Fire.	
Final set of drills containing no less than three continuous drills conducted by and to the satisfaction of the MSEO.	

Note: Signatures required for all lines not greyed.

APPENDIX 2

ENGINEERING WATCHKEEPER

QUALIFICATION BOARD ASSESSMENT GUIDE

ORAL EXAMINATION BOARD PROCEDURE

1. The Oral Examination Board conducted at the end of the Engineering Watchkeeper training package will measure the candidate's understanding of the material studied and the engineering systems they have learned to this point. In order to pass the board, the candidate must receive a mark of at least 60% with the first question having a Pass/Fail criteria for the board (i.e. a failure of this question results in a failure of the board). Board length will be set at a maximum of two (2) hours. Board dates and times will be coordinated by the ship's MSEO. The board shall be composed of the Ship's MSEO (as board Chair), the Chief of the MSE Department, a PO2/PO1 Mechanical MAR TECH, and a PO2/PO1 Electrical MAR TECH.

2. During the board the candidate shall not have access to any references, and will be supplied with the drawing (supplied by NFS(P)). The board shall consist of four (4) questions. The first question will be a description of the system drawing. The candidate will be informed which diagram they are required to present 30 minutes prior to the board to allow the candidate to prepare. The board will then take a break to determine if the candidate has successfully challenged the first question prior to carrying on with the rest of the board. The remaining questions shall consist of a watchkeeping scenario, a safety/environmental policy/considerations, and a question based on general technical knowledge. The questions will be prepared from the question bank provided by NFS(P).

3. In the event of a board failure or training difficulties, refer to the MAR TECH Engineering Watchkeeper QSP.

4. Board Format (suggested timings for questions noted below)

Q1: Engineering Drawing – describe a provided sketch (P/F)
Q2: Scenario Based Watchkeeper Problem (candidate will have time to put plan together)
Q3: Safety / Environmental Policy and Considerations
Q4: General Technical Knowledge

1. ENGINEERING DRAWING (drawing supplied) (45 min)

The candidate is provided with a drawing 30 minutes before the Board. The drawing is of a system with parameters/other details removed. The drawings to be used for this question are the following:

a. Propulsion Diesel Engine air flow;

b. Gas Turbine air flow;

c. Electrical Power Generation and Distribution;

d. Shaftline;

e. Main Gearing;

f. CRPP;

g. Steering;

h. Fuel Oil Services;

i. Main Lube Oil; and

j. IPMS.

The examination board members will have a copy of the drawing with all relevant information provided. The candidate must explain the following to the Examining Board:

a. System Description to include:

(1) Overview of system; and

(2) Purpose of system.

b. System Operation to include:

(1) Normal seagoing configuration; and

(2) Special operating states.

c. Component Description:

(1) Brief purpose and function of each component from the perspective of the Engineering Watchkeeper (not to be treated as a legacy technician board).

d. Parameters

(1) Normal operating ranges; and

(2) Max/min values and warning/alarm points.

e. Direction of flow for:

(1) Fluids; and

(2) Energy, rotation or other mechanisms, as appropriate.

f. Abnormal operation:

(1) Alternate modes in response to a failure or breakdown and their limitations; and

(2) Considerations to the engineering plant in response to any abnormal mode.

g. Emergency Operation

2. SCENARIO BASED (30 min)

The candidate will be given an engineering watchkeeping scenario and will be asked to explain how they would react. The scenario could be taken from any type of watchkeeping incident such as a failure of an engine to start, flash-up procedures or emergencies. Reactions to the scenario are to include:

a. watchkeeper actions;

b. listing personnel/technical resources available to assist; and

c. briefs to Senior Watchkeeper/CHOD/MSEO.

The candidate will be given time to compile their responses prior to responding to the board members.

3. SAFETY / ENVIRONMENTAL POLICY (15 min)

The candidate will be given a question related to workplace safety and environmental policies IAW Safety and Environmental Management System (SEMS) Halifax Class.

4. GENERAL TECHNICAL (Stability and liquid handling or General Watchkeeping) (30 min)

a. STABILITY AND LIQUID HANDLING

The candidate will be given a very basic stability question (ie changes to fuel schedule, difference in fresh water tank levels), to identify the apparent problems this may cause. This question is intended to ensure the watchkeeper is faced with potential changes to stability that can happen during a watch. Follow on aspects can include providing general recommendations and when to call for assistance in correcting the ship's stability.

b. GENERAL WATCHKEEPING

The candidate will be given a basic watchkeeping question (e.g. rounds to be conducted, flash-up procedure, etc), to identify the actions the watchkeeper should take. This question is intended to determine the watchkeeper's general knowledge. This may include general recommendations and when to call for assistance

Copy _____ of 4

Engineering Watchkeeper – Qualification Board Questions

Candidate:	Ship:	Date:
MSEO:	Chief of Department:	PO2/PO1 MAR TECH Mechanical
PO2/PO1 MAR TECH Electrical		

INSTRUCTIONS

Questions are scored out of 10 pts; 0-8 pts for knowledge / understanding and 0-2 pts for presentation / conduct. A minimum of 60% is required for each of the first two questions in order for the board to carry on past the second question.

KNOWLEDGE / UNDERSTANDING:

0 No knowledge. Incapable of answering main part of the question, including where the system/equipment is located, what it looks like or how to begin even researching the topic.

1 Extremely low knowledge. Incapable of answering main part of the question other than where the system/equipment is located, what it looks like or knows where/who to go to in order to obtain more information.

2 Very low knowledge. Continuous assistance was required to answer the entire main question and answers were insufficient in detail.

3 Low knowledge. Continuous assistance was required to answer half or fewer of the main question and answers were insufficient in detail.

4 Marginal knowledge. Frequent assistance was required to answer the main question in acceptable detail. Frequent assistance was required to answer follow up questions and answers were insufficient in detail.

5 Adequate knowledge. Minimal assistance was required to answer the main question in acceptable detail. Frequent assistance was required to answer follow up questions and answers were insufficient in detail.

6 Proficient knowledge. Minimal assistance was required to answer the main question and follow up questions in acceptable detail.

7 Strong knowledge. No assistance was required to answer the main question and follow-up questions in acceptable detail.

8 Exemplary knowledge. Answered main question completely and elaborated in detail on specific aspects such that follow-up questions were unnecessary.

PRESENTATION / CONDUCT:

0 Displayed no confidence or professionalism with board members. Answers/diagrams were vague, unorganized and non-committal.

0.5 Displayed poor confidence or professionalism with board members. Answers/diagrams were partially vague and partially unorganized.

1 Displayed confidence and professionalism with board members. Answers/diagrams were organized.

1.5 Displayed strong confidence and professionalism with board members. Answers/diagrams were organized well.

2 Displayed exemplary confidence and professionalism with board members. Answers/diagrams were highly organized.

Engineering Watchkeeper QUALIFICATION BOARD ASSESSMENT SHEET		
Candidate:	Ship:	Date:

NOTE: Intent is for each assessor to ask one question per section.

Question		Content (0-8)	Conduct (0-2)	Score (S) (0-10)	Comments	
System Drawing	1					
Watchkeeping Scenario	2					
Safety / Environmental Policy and Considerations	3					
General Technical (Stability or General Watchkeeping)	4					
					Total =	
					RESULT (circle one) =	PASS / FAIL

Board Member Comments:

Board Member ID (Mandatory):

Name: _____ Signature:_____

BOARD Average Score	
AVERAGE: [(1)___ + (2)___ + (3)___ +(4)___] / 4 =	

MAR TECH Engineering Watchkeeper Qualification Board Checklist

Candidate: _____ **Board date:** _____

Oral Board Prep:
- **a.** OJPR complete and reviewed
- **b.** Board questions prepared
- **c.** Score sheets prepared
- **d.** Watchkeeping Qualification prepared

Post Board Action:
- **a.** Score sheets signed, corrected and averaged
- **b.** Watchkeeping Qualification signed if member passed
- **c.** Successful completion message drafted and released

COMPOSITION OF MAR TECH ENGINEERING WATCHKEEPER BOARD

Each board exam is to include 8 questions, of which only 4 are used, one per type. Where multiple questions are provided, the board may decide which to use prior to the board start.
The question categories are laid out in the chart below.
A board composition shall be IAW this QSP.
The first question will be asked by the Board Chair.
The second question will be asked by the Chief of Department.
The remaining questions will be asked by the other board members to be determined by the Board Chairperson beforehand.

Question types	Quantity
Engineering problem - drawing	1
Watchkeeper Scenario	3
Safety / Environmental Policy and Considerations	2
General Technical Knowledge (Stability & Liquid Handling OR General Watchkeeping)	2 (1 stability, 1 general)

ANNEX F – ACRONYMS AND ABBREVIATIONS

Acronym and Abbreviation	Term or Definition
CBRN	Chemical Biological Radiological Nuclear
DG	Diesel Generator
EHM	Electronic Health Monitor
GLM	General Load Monitor
GT	Gas Turbine
HVAC	Heating Ventilation Air Conditioning
IMO	International Maritime Organization
IPMS	Integrated Platform Monitoring System
LCC	Local Control Console
LCP	Local Control Panel
LLMS	Liquid Level Management System
LOP	Local Operating Panel
MCC	Machinery Control Console
MDE	Main Diesel Engine
NEM	Naval Engineering Manual
PCOT	Propulsion Control Operator Trainer
PT	Power Turbine
PTIT	Power Turbine Inlet Temperature
SSS	Synchronous Self Shifting

ANNEX G - SUGGESTED SEQUENCES AND TIMINGS

1. The Engineering Watchkeeper course is broken into four POs, each of which contain EOs related to each other for ease of training and subject material. The suggested sequence for delivering the training is as follows with the OJPR completed last:

   a. POs 1 through 3 must be completed in numerical order as follows:

PO 001: Explain the operation of engineering systems					
	EO	**EC**	**PC**	**Multiplier**	
EO 001.01	350		200	1	
EO 001.02	1050				
EO 001.03	300				
EO 001.04	350				
EO 001.05	300				
EO 001.06	550				
EO 001.07	800				
Subtotals:	3700	0	200		
Delivery Method:		**F2F**	**Total (Mins):**		3900
			Total (Days):		9.75

PO 002: Operate IPMS console					
	EO	**EC**	**PC**	**Multiplier**	
EO 002.01	450		100	1	
EO 002.02	440				
EO 002.03	355				
EO 002.04	60				
EO 002.05	250				
EO 002.06	650				
Subtotals:	2205	0	100		
Delivery Method:		**F2F**	**Total (Mins):**		2305
			Total (Days):		5.76

PO 003: Investigate engineering systems				
	EO	**EC**	**PC**	**Multiplier**
EO 003.01	440			1
EO 003.02	300			
EO 003.03	75			
EO 003.04	150			
EO 003.05	350			
Subtotals:	1315			
Delivery Method:		**F2F**	**Total (Mins):**	1315
			Total (Days):	3.29

PO 004: Respond to Alarms and Emergencies				
	EO	**EC**	**PC**	**Multiplier**
EO 004.01	625	50	OJPR	12
EO 004.02	200	50		
EO 004.03	300	150		
EO 004.04	140	150		
Subtotals:	1265	400	0	4800
Delivery Method:		**F2F**	**Total (Mins):**	6065
			Total (Days):	15.16

Note: The timings presented above do not contain any accounting of TAT, BAT or PT. The column labeled as “multiplier” indicates the number of times that testing runs must be conducted to capture all 12 students once.

2. The following table provides the expected course duration information with the maximum trainees considered (12):

	EO	**EC**	**PC**	**TAT**	**BAT**	**PT**	**TTT**	**Divisions**
Minutes	4835	3000	100	120	120	720	600	120
Days	12.09	7.5	0.25	0.3	0.3	1.8	1.5	0.3

Total	**Time**
Days	24.04
Weeks	4.808
Months	1.202

3. The TE is expected to review all ECs and PCs to confirm that they can accomplish the required testing within the allocated times. If the TE cannot achieve the aim, two options

are to be considered: purchasing required training material to meet the timings or increase the time allocated to the course to achieve the aim.

ANNEX H - TRAINING RESOURCES

1. The following is a list of the resources identified as being required to conduct the Marine Technician Engineering Watchkeeper Course. Some of the resources listed currently exist at the Naval Fleet Schools; however, there are new resources that will be required to maximize the efficiency of training delivery and testing. Purchasing the new items will increase the quality of the student graduating from the training, and significantly reduce the number of required training days ultimately saving time and money over the long-term.

2. The following is a list of classroom requirements:

   a. One large classroom to accommodate maximum number of candidates which include:

      (1) Individual candidate's desks c/w chair;

      (2) Smart Board;

      (3) Dry erase board;

      (4) Projector (power point capable) & screen;

      (5) BluRay Player & TV;

      (6) Podium;

      (7) DWAN / LAN connections;

      (8) Laptops/PCs;

      (9) Calculators;

      (10) PPS; and

      (11) Student handouts and references.

   b. PCOT to accommodate maximum number of trainees.

      (1) Current course loading capacity is 10 trainees. IOT achieve maximum capacity of 12 trainees two additional laptop workstations should be purchased for PCOT.

   c. MCR trainer.

ANNEX I – COSTING

	UNIT	PRICE	QTY	TOTAL
Supplies (STUDENTS)				
1" Binder	EA	$9.39	12	$112.68
Eraser	EA	$2.15	12	$25.8
Blue Pen	EA	$0.69	12	$8.28
Pencil	BOX	$6.39	1	$6.39
Lined Paper	PKG	$9.99	12	$119.88
Highlighter	BOX	$21.48	1	$21.48
PCOT Terminals	EA	TBD	2	TBD
SUBTOTAL				$294.51
Supplies (INSTRUCTOR)				
Blue Pen	EA	$0.69	2	$1.38
Red Pen	EA	$0.69	2	$1.38
Pencils	BOX	$6.39	1	$6.39
Post It Notes	PKG	$5.89	4	$23.56
Writing Pad 8.5" x 11"	EA	$4.79	4	$19.16
Whiteboard Markers	PKG	$6.99	1	$6.99
Whiteboard Eraser	EA	$11.99	1	$11.99
Paper Clips	BOX	$1.09	4	$4.36
Sharpie	EA	$3.69	2	$7.38
CDROM	PKG	$18.33	1	$18.33
SUBTOTAL				$82.59
*This table does not include costing for travel and R&Q				

ANNEX J- COURSE HISTORY

The USQ Engineering Watchkeeper QSP was developed from the legacy MCC QSP material and input from representatives from ER, ET, and HT legacy MOSIDs. This course is a follow-on from the Marine Technician USQ S1 Mechanical Course, and is a prerequisite for employment as an Engineering Watchkeeper on board HFX class platforms.

ANNEX K – ASSESSMENT PLAN

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria	Impact
EC001.01		Explain IPMS	Homework	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 001.02		Explain Ancillary and Auxiliary Systems	Homework	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 001.03		Explain PG&D	Homework	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
PC 001.03		Assessment EC 001.01 through EC 001.03	Written test	60%	First PC Fail results in a retest Second PC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
EC 001.04		Explain Shafting Arrangements	Homework	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 001.05		Explain Main Gearing	Homework	Nil	Trainee issued Trainee Experiencing Difficulty (TED)

				and provided additional training/instruction.
EC 001.06	Explain PDE	Homework	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 001.07	Explain GT	Homework	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
PC 001.07	Assessment of EC 001.04 through EC 001.07	Written test	60%	First PC Fail results in a retest Second PC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
EC 002.01	To assess the trainees ability to operate machinery console.	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 002.02	To assess the trainees ability to operate auxiliary systems.	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 002.03	To assess the trainees ability to lead collective operation of	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided

		ancillary systems.			additional training/instruction.
EC 002.04		To assess the trainees ability to operate PG & D.	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 002.05		To assess the trainees ability to operate gearing and shafting.	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EO 002.06		To assess the trainees ability to operate main propulsion.	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
PC 002	EO 002.01 – 002.06	To assess the trainee's ability to operate IPMS console.	Practical Assessment	7/11, no critical failures as per PC checklist.	First PC Fail results in a retest Second PC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
EO 002.05		To assess the trainees ability to investigate Main Propulsion Systems.	Developmental Observations	Nil	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.

EC 004.01	EO 003.01 – 003.02, 004.01	To assess the trainees ability to respond to auxiliary and ancillary alarms.	Practical Assessment	76 Points	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction
EC 004.02	EO 003.03 004.02	To assess the trainees ability to respond to PG & D alarms.	Practical Assessment	TBD	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
EC 004.04	EO 004.03 and 004.04	To assess the trainee's ability to respond to ship emergencies.	Practical Assessment.	76 Points	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
OJPR	PO 002 003 and 004	To assess the trainee's ability to investigate engineering systems and respond to alarms and emergencies.	Practical Assessment with final oral board.	Completion of OJPR package.K,	First Board Fail results in a second board with a training extension Second Board failure results in a TRB IAW NPTORD. Remedial training/counselling as required.

ANNEX L – TECHNOLOGY ENABLED LEARNING

1. This course will take advantage of pre-existing TEL in the form of simulators, PCOT, and MCR trainer located at NFS(A) and NFS(P).

2. With limited real estate available at both NFS(A) and NFS(P), consideration for a Multi-Use Reconfigurable Trainer (MRT) would allow for training for the RCNs current and future multiple platforms.